

Coverage Substitution After Job Loss: How the ACA Altered Post-Displacement Insurance Trajectories

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ABSTRACT

Did the Affordable Care Act soften the post-displacement coverage shock that follows from the U.S. employer-tied insurance system? The 2014 introduction of community-rated Marketplace plans and the Medicaid expansion gave separated workers non-employer alternatives that the pre-ACA non-group market did not offer, and this paper measures whether those alternatives changed what happens to insurance coverage and out-of-pocket premium burden when workers lose their jobs involuntarily. Using the Medical Expenditure Panel Survey 2004–2018 and a matched event-study design adapted from [Schaller and Zerpa \(2026\)](#), I estimate era-specific displacement effects on a pre-ACA cohort and a post-ACA cohort and compare them. Post-ACA workers lose meaningfully less coverage than their pre-ACA counterparts: the ACA absorbs roughly one-quarter of the two-month post-displacement coverage loss, with the gains concentrated in the income bands the subsidy and expansion thresholds were designed to reach. The cost story has two parts that move in opposite directions. After displacement, more workers in the post-ACA cohort end up on a plan that requires them to pay a premium (consistent with substitution onto non-employer private plans, which are not employer-paid by construction), but among those who pay any premium, more workers spend over 10% of their household income on it (the high-burden share roughly doubles between eras). Marketplace coverage is real insurance, but it is not cheap insurance. A latent-class analysis on baseline job characteristics adds a third layer: coverage gains concentrate in a high-coverage Finance and Professional class, while three precarious-work classes show no coverage gains, with what gains there are operating through the public-insurance margin rather than through Marketplace substitution. The post-2014 architecture reaches the populations its thresholds were designed to reach, leaves a meaningful within-payer cost burden on the workers who do take up the new coverage, and reaches the precarious-work population unevenly.

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1 Introduction

Health insurance in the United States is bundled with employment. This produces two related problems. The first is *job lock*: workers stay in jobs they would otherwise leave because leaving means losing coverage. The second is its mirror: when a worker loses a job involuntarily, the worker also loses the insurance that came with it. Both come from the same feature of the system, the link between coverage and the job, but they sit on opposite sides of the separation event. The job-lock literature has mostly studied the first problem: [Madrian \(1994\)](#) provided the first credible empirical evidence that employer-sponsored insurance (ESI) reduces voluntary turnover by roughly 25 percent, and [Gruber and Madrian \(2002\)](#) synthesize a decade of follow-up work confirming the result. The friction is widely recognized: in nationally representative survey data, roughly 90% of Americans agree that workers stay in jobs they would otherwise leave because of the fear of losing employer-tied coverage ([Ferrario and Stantcheva, 2025](#)). The Affordable Care Act of 2010, by creating Marketplace plans with income-scaled subsidies and expanding Medicaid, was expected to ease both problems. This paper studies the second one: did the ACA measurably reduce the coverage shock that follows involuntary displacement? The recent empirical job-lock literature, including [Schaller and Stevens \(2015\)](#) and [Schaller and Zerpa \(2026\)](#), has moved its focus from voluntary mobility (the original Madrian framing) toward involuntary displacement, where post-ACA policy variation provides cleaner identification than the cross-sectional comparisons that dominated the 1990s literature. The analysis here is reduced-form: it tests one observable consequence of the ESI tie and measures what the ACA changed, and complements rather than substitutes for the structural welfare estimates in [Aizawa and Fang \(2020\)](#) and [Fang and Shephard \(2019\)](#). [Dey and Flinn \(2005\)](#) caution that reduced-form patterns of this kind do not automatically translate into welfare losses without an equilibrium model, and the same caution applies to the displacement-margin estimates here.

Two empirical questions follow. First, did the ACA measurably reduce the coverage loss that follows involuntary displacement? Second, did its coverage gains arrive cheaply, or did the displaced workers who took up Marketplace coverage face meaningful out-of-pocket cost? Using the Medical Expenditure Panel Survey (MEPS) 2004–2018, I identify involuntary separations among workers aged 26–59 and match each displaced worker to a non-displaced control on calendar month, gender, region, and baseline income group. I estimate a propensity score within these exact cells, use inverse-odds reweighting to align the pre- and post-ACA samples on baseline demographics, and recover dynamic event-time profiles separately for the pre-ACA period (2004–2013, excluding the Great Recession) and the post-ACA period (2014–2018). The procedure follows [Schaller and Zerpa \(2026\)](#) closely, adapted from the Survey of Income and Program Participation (SIPP) to the MEPS monthly panel structure.

Three findings structure the paper. First, *post-ACA displaced workers lose meaningfully less coverage than their pre-ACA counterparts*. The ACA absorbs roughly one-quarter of the coverage loss two months after displacement and one-third of the loss four months out. The reduction comes through two channels: more displaced workers end up on a non-employer private plan (predominantly Marketplace) rather than no plan at all, and a modest increase in public-coverage take-up. The pooled mitigation estimates point in the policy-relevant direction at every post-displacement horizon but do not individually clear conventional significance thresholds; the heterogeneity results in the third finding explain why. The design itself passes standard internal checks: pre-displacement insurance trends are indistinguishable from zero in both eras; panel-level estimates show a clear break at 2014 rather than a gradual trend; and a test that exploits baseline ESI exposure shows that workers covered through their own employer plan at baseline lose substantial coverage after displacement, while workers without an employer offer at all show essentially no displacement-induced coverage loss, since their coverage was not tied to the job in the first place (§5.2).

Second, *the cost story has two parts that move in opposite directions, and a single-stage burden regression averages them out*. Displacement moves a worker across two transitions at once: from being privately insured to potentially uninsured, and within the privately-insured group, from a plan whose premium the employer paid to a plan that requires worker contributions. Averaging both transitions in one regression produces a near-zero summary that hides both. Following [Goldman et al. \(2018\)](#), I split the burden outcome into two stages: whether the worker pays any premium at all (Stage 1A), and how large the premium-to-income ratio is conditional on paying (Stage 2). The two stages move in opposite directions. Stage 1A shows positive ACA mitigation: more displaced post-ACA workers end up on a premium-paying non-employer plan rather than dropping out of private coverage entirely. Stage 2 shows the opposite: among displaced workers who pay premiums, the post-ACA group is more likely to be at the high-burden end of the distribution than the pre-ACA group was. The substantive reading is that Marketplace coverage is real insurance for the displaced but it is not cheap insurance, and the coverage gain and the cost burden should be tracked as two separate policy outcomes.

Third, *the average mitigation hides large differences across populations that are informative for policy*. The mitigation is concentrated in the income range where the ACA's subsidies and Medicaid expansion target most directly: the Low Income bin running from 100% to 199% of the federal poverty line, which spans the Medicaid-expansion threshold (138% FPL) and the lower portion of the Marketplace-subsidy range. A latent class analysis on baseline job characteristics identifies four worker types and shows a sharper pattern: the post-ACA coverage gains are concentrated in the high-coverage Finance and Professional class and come through workers ending up on premium-paying non-employer plans rather than dropping coverage. The three precarious

classes show no coverage gains. The Trade and Sales class in particular has a downward post-ACA cohort trajectory at every horizon, including before displacement, which I read as a general post-2014 coverage decline among trade and sales workers rather than a displacement-specific ACA effect. Joint tests across the four worker types confirm that the policy reaches them unequally on coverage, even though the cost burden conditional on paying any premium is similar across worker types. Within the precarious classes, the small coverage gains that do appear come through public insurance rather than through Marketplace plans, which suggests take-up rather than eligibility is the binding constraint for these workers, while income-targeted provisions reach the Good-job and middle-income workers as designed.

This paper contributes to three literatures. First, it extends [Schaller and Zerpa \(2026\)](#)’s post-displacement coverage-shock evidence on the Survey of Income and Program Participation to a parallel matched event-study on the MEPS, separating the coverage outcome from a Marketplace-era affordability outcome via a two-part hurdle decomposition. Second, it provides reduced-form evidence on one observable implication of the ESI dependency that the job-lock literature identifies ([Madrian, 1994](#); [Gruber and Madrian, 2002](#)), namely that severing employment severs coverage; this evidence is complementary to the structural welfare estimates in [Aizawa and Fang \(2020\)](#) and [Fang and Shephard \(2019\)](#), and the baseline-ESI-exposure test (§5.2) sharpens the identification by showing that post-displacement coverage loss is concentrated in workers who actually had ESI to lose. Third, applying the latent-class methodology of [Brauner et al. \(2019\)](#) to baseline job-quality indicators connects the displacement and ACA literatures to the precarious-work literature ([Aboulafia and Schneider, 2025](#)), showing that the workers who benefit most from the ACA and those left behind are distinct, empirically separable occupational groups (§5.4).

The paper proceeds as follows. Section 2 summarizes the relevant ACA provisions and prior literature. Section 3 describes MEPS and the construction of the analysis file. Section 4 details the empirical strategy. Section 5 presents the main coverage, burden, and heterogeneity results along with validation and robustness checks. Section 6 discusses what the design does and does not identify, limitations, mechanism, and policy implications. Section 7 concludes. Section 8 presents all tables and figures.

2 Background

This section reviews the institutional setting, the policy provisions the paper studies, and the prior empirical evidence the design builds on. Section 2.1 describes how employer-sponsored insurance generates job lock and post-displacement coverage loss. Section 2.2 summarizes the two main ACA provisions that bear on what happens to workers when they lose employer coverage. Section 2.3 reviews the displacement and ACA literatures the paper extends.

2.1 The ESI–employment tie and its frictions

In the United States, working-age adults are insured primarily through the private market, and within the private market, primarily through their employer. Employer-sponsored insurance (ESI) is the dominant coverage source for this age group: the employer purchases a group plan and the worker enrolls in it. Public programs are smaller for working-age adults and concentrated in specific groups: Medicare for the elderly, and Medicaid for low-income adults and families. Figure 1 plots the monthly coverage rates on the working-age MEPS panel and shows that ESI is the dominant private-coverage channel throughout the sample period. For working-age adults outside these groups, leaving an ESI-providing job typically means losing coverage. This arrangement creates a friction known as *job lock*: a worker who would otherwise leave a job hesitates because leaving means losing the insurance that came with it.

Three features of the pre-ACA market made the friction sharp. The standard bridge between jobs, the Consolidated Omnibus Budget Reconciliation Act provision (COBRA), allows a separated worker to continue the employer plan for up to eighteen months but requires paying the full employer premium plus an administrative fee, which is unaffordable for most workers during unemployment. The non-group market that existed before 2014 used medical underwriting, meaning insurers could deny coverage, charge unaffordable premiums, or exclude pre-existing conditions for workers with serious health histories. New ESI plans typically imposed waiting periods and pre-existing-condition exclusions for several months. Workers with the highest expected medical needs (those with chronic conditions, dependents, pregnant spouses, and those near but not yet eligible for Medicare) faced the largest cost from leaving a job that provides coverage (Madrian, 1994; Gruber and Madrian, 2002). Grossman (1972) clarifies why this matters beyond the immediate dollar cost: in his framework health is a stock that depreciates with age and is replenished through medical care, so ESI lowers the price of that investment and losing coverage reduces the worker’s optimal health stock.

The friction operates on two distinct margins. On the *voluntary mobility* margin, workers stay in jobs they would otherwise leave; Madrian (1994) provided the foundational identification using a difference-in-differences across whether a worker held ESI and whether the worker had high expected medical expenses, finding that voluntary annual turnover is roughly one-quarter lower for ESI-covered workers. On the *involuntary displacement* margin, workers who lose jobs lose coverage at the same time. Garthwaite et al. (2014) use the 2005 TennCare disenrollment in Tennessee to make the symmetric point in reverse: when public coverage disappeared, adults entered the labor force in measurable numbers to obtain ESI. The two margins are not mutually exclusive (a single worker can experience both), but they require different identification strategies. A third margin, which this paper does not directly test, is dynamic underinvestment in worker health: because workers turn over, neither the current employer-employee pair nor the worker individually inter-

nalizes the full long-run return to investing in worker health, so preventive-care investments whose payoff accrues to a future employer are systematically underprovided (Fang and Gavazza, 2011). Whether the reduced-form turnover differential implies a welfare loss remains debated: Dey and Flinn (2005), in a search-and-bargaining equilibrium model, argue that most of the longer match duration of insured jobs reflects the productive role of insurance in the match rather than workers being trapped, and conclude that ESI “does not lead to any serious inefficiencies in mobility decisions.” The empirical coverage trajectories this paper documents should be read as facts about post-displacement insurance, not in themselves welfare estimates; translating coverage facts into welfare requires a structural model that takes a stand on whether the ESI–wage bundle is bargained or imposed.

2.2 The ACA’s coverage architecture

The Affordable Care Act of 2010 introduced two main provisions, both effective January 2014, that bear directly on what happens to workers when they lose employer coverage (Ortaliza et al., 2025). Throughout this paper, household income is expressed relative to the federal poverty line (FPL), an annual income threshold published by the U.S. Department of Health and Human Services that varies by household size; for example, the 2014 FPL was approximately \$11,670 for a single adult and \$23,850 for a family of four.

The first provision, the *Health Insurance Marketplaces*, are federally and state-operated exchanges through which individuals not covered by an employer can purchase non-group plans at community-rated prices. Community rating means insurers cannot deny coverage on the basis of pre-existing conditions and cannot vary premiums by health status, but only by age, geography, family size, and tobacco use. *Income-scaled premium tax credits* reduce the monthly premium for households between 100% and 400% of FPL, with additional cost-sharing reduction subsidies for households between 100% and 250% of FPL. This describes the original ACA architecture, operative across the post-2014 sample window; subsequent legislation (the 2021 American Rescue Plan and 2022 Inflation Reduction Act) raised the contribution generosity and removed the 400% upper bound, but those changes lie outside the study window (see Section 6). The second provision is the *Medicaid expansion*, which extended Medicaid eligibility to all adults in participating states with household income below 138% of FPL, regardless of categorical qualifications previously required for adult Medicaid. Because the 2012 Supreme Court decision in *NFIB v. Sebelius* made the Medicaid expansion optional for states, take-up has been uneven. The main analysis treats the 2014 cutoff as a national intent-to-treat shock rather than estimating state-level rollout effects.

A separate dependent-coverage provision, in force since September 2010, allows adults under age 26 to remain on a parent’s employer plan; this paper restricts the analysis sample to adults aged 26 and over to isolate the post-2014 Marketplace and Medicaid channels.

Both Marketplace and Medicaid implement the same basic policy idea. By making non-employer coverage widely available, they should reduce the welfare cost of involuntary separation that the [Madrian \(1994\)](#) framework predicts. The Marketplace channel matters for middle-income workers whose income falls but remains above the Medicaid threshold; the Medicaid channel matters for lower-income workers whose income falls below 138% of FPL after job loss. Together they implement a *partial decoupling* of insurance from employment: non-employer alternatives become widely available, but employer coverage remains the modal source for working-age adults.

2.3 Prior empirical evidence and the contribution of this paper

The displacement-event-study tradition begins with [Jacobson et al. \(1993\)](#), who document large and persistent earnings losses for high-tenure workers losing jobs in mass layoffs. [Schaller and Stevens \(2015\)](#) extend this design to short-run insurance and health outcomes in the pre-ACA period, showing that involuntary displacement caused immediate insurance loss, reduced utilization for chronic conditions, and worsened health. The closest pre-ACA precedent for an ACA-style policy intervention is [Gruber and Madrian \(1997\)](#), who used state continuation-of-coverage mandates as identifying variation and found that even short continuation mandates measurably restored mobility (a 6.7% increase in coverage probability after separation) and lengthened non-employment durations during productive job search.

[Schaller and Zerpa \(2026\)](#) provide the closest comparator to this paper. Using SIPP panels 2004, 2008, and 2014 (covering 2004–2016) with a matched event-study design and cross-era reweighting, they find that post-displacement coverage losses are meaningfully smaller after the ACA than before, with faster recovery and gains concentrated among middle-income workers. They identify two operating channels: post-ACA workers enter displacement with more public coverage already in place, and conditional on displacement they lose less private coverage. Their result is the empirical anchor this paper builds on, and their identification strategy is the design template I adopt and extend to MEPS.

A parallel structural literature targets ACA welfare quantities. [Aizawa and Fang \(2020\)](#) estimate a search model and use it to study the ACA, finding that the income-based premium subsidies are the operative provision: without them, healthy enrollees stop buying insurance on the new exchanges, prices rise sharply from adverse selection, and the marketplaces collapse. The reduced-form approach in this paper is complementary: the patterns documented here are the moments structural welfare exercises target.

On the population-level burden margin, [Goldman et al. \(2018\)](#) document on MEPS 2012–2015 data that mean post-ACA premium contributions rose, driven primarily by households above 400% of FPL who are not eligible for premium subsidies, while out-of-pocket medical spending fell sharply across the three lower-income groups. This paper extends that pattern to the displacement-

specific margin: how do post-ACA coverage and burden trajectories evolve at the moment workers cross from ESI-protected to displacement-vulnerable status?

Finally, displacement consequences vary across worker types in ways household income alone does not capture. [Brauner et al. \(2019\)](#) provide the methodological template, using latent class analysis (LCA) to recover six empirically distinct working-time-control profiles in a German workforce and finding that worker types with high demands and low control report meaningfully worse health and work-life balance. [Aboulafia and Schneider \(2025\)](#) document a U.S.-specific take-up barrier: the ACA’s employer-mandate exemptions, namely franchise structure as a proxy for the 50-employee firm-size threshold, part-time hours below the 30-hour weekly cutoff, and short-tenure exemptions allowing 90-day waiting periods, systematically reduce the probability of an employer offer for low-wage hourly workers, and the affected workers map onto the precarious worker types this paper recovers via LCA. Together these motivate the multi-dimensional heterogeneity analysis in Section 5.4.

3 Data

This section describes the Medical Expenditure Panel Survey (MEPS) data the paper uses and the construction of the analysis sample. Section 3.1 introduces the survey and the three public-use files that supply the variables. Section 3.2 describes the sample restrictions, including the involuntary-displacement treatment and the matched-control construction. Section 3.3 defines the five outcome variables and discloses the construction choices that affect the headline estimates. Section 3.4 reports the descriptive statistics.

3.1 Survey structure and sources

The Medical Expenditure Panel Survey Household Component (MEPS-HC) is a panel survey of the U.S. civilian noninstitutionalized population, fielded annually by the Agency for Healthcare Research and Quality (AHRQ) since 1996 ([Cohen, 1997](#); [Chowdhury et al., 2019](#)). Each panel covers two consecutive calendar years across five in-person interview rounds, with monthly insurance status, employment, and out-of-pocket premiums collected each round. For any given calendar year two overlapping panels are simultaneously in the field, and AHRQ combines them into the annual Full-Year Consolidated Public Use File. I pool the annual files for calendar years 2004 through 2018, a window that spans both sides of the ACA’s January 2014 implementation date and includes the Great Recession of 2008 to 2010, which I treat as a separate period throughout.

The analysis file is built from three MEPS-HC public-use files. The Full-Year Consolidated file provides demographic and socioeconomic information at the person-year level, including the monthly insurance-coverage indicators that yield the three coverage outcomes (any, private, pub-

lic). The Jobs file provides the employment history at the person-round-job level, including industry, occupation, hours, earnings, employer-sponsored insurance offer, non-wage benefit indicators, and the reason-for-leaving codes that identify involuntary separations. The Person Round Plan file provides the out-of-pocket premium amount, defined only for person-months with private coverage. The three files are linked at the person level by a harmonized identifier and at the job level by a job index. Exact AHRQ release numbers are in Appendix A.

3.2 Sample: working-age adults facing involuntary separation

The analysis sample is restricted to adults aged 26 to 59 with complete demographic information. The lower cutoff of 26 excludes adults eligible for parental coverage under the ACA’s dependent-coverage provision, which operates through a different mechanism than the post-2014 Marketplace and Medicaid channels this paper studies; [Dillender \(2014\)](#) documents the dependent-coverage provision specifically, and the ACA’s coverage architecture is reviewed in Section 2.2. The upper cutoff of 59 keeps the sample away from Medicare eligibility at 65 and from near-retirement transitions.

A central design choice is to restrict the treatment to *involuntary* job separation (layoffs, firings, and business closures) rather than all job endings. Voluntary quits are not independent of the worker’s insurance situation. Workers who quit disproportionately already hold spousal coverage, have a new offer in hand, or have savings to bridge a gap, all of which would bias the estimated coverage consequences of job loss toward zero. Involuntary separations originate with the employer or with a business failure, and their timing is plausibly orthogonal to the worker’s insurance position at baseline. Following [Schaller and Zerpa \(2026\)](#), displaced workers must additionally satisfy three conditions: continuous employment for at least six months before separation, providing a stable pre-period; out of employment for at least one month in the three months following separation, ruling out nominal separations followed immediately by a new job; and not self-employed at baseline, since the employer-tied job-lock mechanism does not apply to workers who buy individual plans directly. Controls are workers continuously employed for at least six months, not self-employed, and never displaced during their panel. I additionally drop workers in Military industry or occupation codes, because active-duty service members are covered by TRICARE, the military health system, which eliminates the private-sector job-lock mechanism this paper studies.

The sample construction proceeds as follows. Starting from all person-month observations in the 2004 to 2018 window, I retain adults aged 26 to 59 with complete demographic information, producing a person-month panel of approximately 5.9 million observations on 301,111 unique respondents. Applying the involuntary-separation codes together with the six-month pre-period, three-month post-period, and not-self-employed conditions leaves 1,983 treated workers eligible for matching. After the propensity-score matching procedure described in Section 4.2, the matched

pool contains 1,889 treated and 1,820 control workers (3,709 persons total, a 95.3% match rate). Dropping the Military cases and a small number of workers with listwise-missing latent-class indicators yields the *analytical sample* of 3,665 persons (1,875 treated, 1,790 controls) used throughout the paper.

3.3 Outcome variables

Variable definitions. The analysis uses five outcome variables, all constructed from MEPS person-month and plan-month files: three person-month coverage indicators and two continuous cost measures, plus two binary indicators derived from the cost measures by thresholding.

Any coverage equals one if the respondent held either private or public coverage in month t .

Private coverage equals one for any private plan, including employer-sponsored insurance, direct-purchase non-group plans, and post-2014 Marketplace plans, grouped together because they share the private-market cost structure. The pooled private indicator is the policy-relevant object: post-displacement, a worker who remains privately insured may do so through any of these channels, and the pre/post-ACA contrast in private coverage captures the net effect of the policy on the non-employer side of the private market regardless of which sub-channel the worker actually uses. The MEPS public-use file does not separately identify Marketplace plans, so Marketplace and other direct-purchase non-group plans are not directly distinguishable in the data. Where the discussion attributes a post-2014 shift in non-employer private coverage to the Marketplace channel, the attribution is interpretive rather than directly measured; the empirical estimand throughout is total private coverage.

Public coverage equals one for Medicaid, Medicare, or other public coverage; Medicaid is dominant in the 26 to 59 age range.

The *out-of-pocket premium* is the dollar amount the policyholder pays each month to keep a private plan in force. It is a premium contribution, not a measure of medical-care spending: deductibles, copays, and coinsurance are excluded. MEPS records this at the plan-month level. By construction the premium is observed only for person-months with private coverage, and is zero for plans whose entire premium is paid by the employer or, in a small number of cases, by a government program.

The *premium burden* is the share of household income devoted to out-of-pocket premium payments at the monthly frequency: $\text{burden}_{i,t} = \text{premium}_{i,t} / Y_{i,t}^{\text{HH}}$, where the household is the MEPS health-insurance eligibility unit (typically an adult, spouse, and unmarried children up to age 18 or 23 if full-time students). The denominator follows [Selden \(2009\)](#) in spirit and is the sum across household members of monthly labor earnings, transfer income (unemployment compensation, public assistance, SSI, and similar), and unearned income (interest, dividends, pensions and IRA withdrawals, Social Security, veterans' payments, and similar); annual income components are

allocated to months following the rules in Appendix B. All dollar series are deflated to constant 2016 dollars using the BLS Consumer Price Index for All Urban Consumers (U.S. Bureau of Labor Statistics, 2024); because the same deflator multiplies numerator and denominator, the burden ratio is invariant to inflation adjustment.

Two derived binary indicators enter the empirical analysis as Stage 1A and Stage 2 outcomes of the two-part hurdle decomposition introduced in Section 4.5. The *premium-paying indicator* $\mathbb{1}\{\text{premium}_{it} > 0\}$ equals one if the worker pays any positive out-of-pocket premium in month t . The *high-burden indicator* $\mathbb{1}\{\text{burden}_{it} > 0.10\}$ equals one if the burden ratio exceeds 10%, a threshold widely applied in the underinsurance literature including Goldman et al. (2018); the deviation from Selden (2009)’s 20% combined-spending threshold is discussed below.

Construction details and disclosures. *Winsorization and floor-binding.* The out-of-pocket premium is winsorized at the 95th percentile of the private-coverage distribution; the burden ratio is winsorized at its 99th percentile within the burden-valid sample. To prevent burden from being unstable when household income is near zero, I bound household income from below at \$1,000 per year in 2016 dollars (about \$83 per month), following Selden (2009, footnote 8). Household-months with income below the floor are flagged *floor-binding* and excluded from the burden-valid sample. Construction details, the floor-binding rate, and alternative-floor sensitivity diagnostics are in Appendix B. *Deviations from Selden (2009).* Three deviations warrant acknowledgment. First, my numerator is premium-only rather than premium-plus-point-of-service spending, because the relevant question for a displaced worker is the price of remaining covered, not unrelated medical events. Second, my denominator is gross household income rather than disposable income, since simulating the federal and state tax burden at the household-month level is beyond the scope of this thesis. Third, my high-burden cutoff is 10% rather than Selden’s 20% on combined spending. Sensitivity at 5% and 15% is reported in the appendix and is consistent with the headline pattern.¹ *Premium-missing person-months.* I keep a missingness indicator for person-months without private coverage and never treat zero and missing as the same; doing so would conflate the coverage margin (whether the worker has private coverage) with the cost margin (how much the worker pays conditional on having it).

3.4 Descriptive statistics

Table 1 reports summary statistics for every variable entering the causal analysis on the full working-age MEPS panel over 2004 to 2018, split by policy era. The era split reveals large

¹Workers’ compensation is the one income source from Selden’s transfer list that I cannot reproduce, because the relevant variable is not released in the public-use file for any year in my 2004 to 2018 sample window. Workers’ compensation is small in absolute size relative to the other transfers, and its omission biases reported burden ratios slightly upward (a smaller denominator) for the small share of households that would have received it.

aggregate shifts. Any-insurance coverage rises from 78.2% pre-ACA to 85.2% post-ACA, and public coverage rises from 35.3% to 43.3%. Private coverage is essentially flat at the population level (49.2% pre-ACA, 49.8% post-ACA), implying that virtually all of the aggregate coverage gain comes through public-program expansion rather than through private substitution. Employer-sponsored insurance offer rates among the employed are also flat across eras, confirming that the ACA did not displace employer-sponsored coverage at the population scale. Mean monthly out-of-pocket premium among privately-insured person-months rises in real terms from \$214 to \$245 (constant 2016 dollars), and the share of those person-months with high burden (above 10%) rises from 16.7% to 18.0%. These are population-level shifts rather than the displacement-specific shifts that motivate the event study; contrasting them with the event-study estimates in Section 5 is one reason the design is needed.

Table 2 reports the parallel summary on the matched analytical sample at baseline. Compared with the full panel, the matched sample skews older (mean age 40, compared with 35 in the full panel), is more often full-time (80.3% in the matched sample, 57% in the full panel), and is meaningfully more covered (any-insurance 67.7% at baseline; employer offer 69.0%, well above the full-panel offer rate of 53%). These differences reflect the six-month tenure requirement and the not-self-employed selection criterion. The cross-era contrast at baseline shows higher coverage in the post-ACA cohort (75.4% in post-ACA, 66.2% in pre-ACA on any insurance), driven primarily by public-program take-up. The treated cohort divides into 1,577 pre-ACA workers and 312 post-ACA workers; the small post-ACA sample limits the precision of post-only and stratified estimates and motivates the borderline-significance language reported in Section 5.

Figures 1 through 4 plot four aggregate time-series trends underlying the design: the monthly insurance coverage rates on the working-age panel (Figure 1, introduced in Section 2.1); the annual rate and count of involuntary job loss (Figure 2, motivating the separate Great Recession era control); the monthly uninsurance rate by baseline poverty category (Figure 3); and the mean and median monthly out-of-pocket premium among privately-insured person-months (Figure 4). Together they provide the population-level context against which the displacement-specific event-study estimates can be read.

4 Empirical Strategy

This section sets out the identification strategy and the estimating equations the paper uses. Section 4.1 states the four empirical questions and the identifying assumption. Sections 4.2 and 4.3 describe the matching procedure and the cross-era reweighting that address the two confounders. Sections 4.4 and 4.5 present the event-study specification and the two-part hurdle decomposition for the burden outcomes. Section 4.6 introduces the latent-class analysis and the triple-difference

specification used in the heterogeneity analysis.

4.1 Objective and identification

I identify the causal effect of involuntary job loss on health insurance coverage and out-of-pocket premium burden, and test whether that effect changed after the ACA’s main provisions came into force in January 2014. “Job loss” throughout refers to separations the worker did not initiate (layoffs, plant closings, and other involuntary-displacement codes described in Section 3), so the event can plausibly be treated as a shock to the worker rather than an outcome of her own decision.

The empirical strategy answers four questions. First, what happens to displaced workers’ insurance coverage and premium burden in each era (Section 5.1)? Second, is the mechanism the ESI dependency the job-lock literature identifies (Section 5.2)? Third, does the post-ACA trajectory differ from the pre-ACA trajectory in the income bands the policy thresholds predict (Section 5.3)? Fourth, do the era-specific trajectories differ across types of workers when the sample is partitioned by an empirical job-quality classification rather than by income alone (Section 5.4)?

Two confounders threaten a naive answer to these questions. First, displaced workers are not a random subset of the working-age population, so a comparison of pre/post-displacement coverage confounds the causal effect of job loss with selection on baseline characteristics. Second, the workers displaced in 2006 are not the workers displaced in 2016, so a cross-era comparison among displaced workers confounds policy with compositional change.

Closely following Schaller and Zerpa (2026), my quasi-experimental strategy exploits the January 2014 ACA implementation as a natural experiment and addresses both confounders in sequence. I match each displaced worker to a non-displaced control on calendar month, sex, region, and baseline income group, plus a propensity-score match within cell (Section 4.2); reweight the pre-ACA matched pairs so their baseline demographics match the post-ACA pairs (Section 4.3); estimate a two-way fixed-effects event study with event time interacted with three mutually exclusive policy eras (Section 4.4); and decompose the burden outcomes into a two-part hurdle that separates extensive and intensive margins (Section 4.5). For the worker-type heterogeneity analysis, a latent-class model on 23 baseline job-quality indicators feeds into per-class and triple-difference event studies (Section 4.6).

The identifying assumption is that, conditional on the matching covariates and the cross-era reweighting, no contemporaneous structural change other than the ACA differentially altered the causal effect of involuntary job loss on insurance coverage between the pre-2014 and post-2014 eras. Under that assumption, the difference $\hat{\beta}_k^{\text{Post}} - \hat{\beta}_k^{\text{Pre}}$, which I call *ACA mitigation* at event time k , is a causal estimate of the policy’s effect on the insurance consequences of displacement.

4.2 Matching

Matching builds a control group that resembles the treated group on observables that predict both displacement and insurance status, and matches *within* narrow cells of calendar time and income so each treated-control pair shares the same macroeconomic moment and Medicaid-eligibility band. Whatever else is happening at that time affects both workers identically and cancels in within-pair comparisons. I implement a two-step procedure following [Schaller and Zerpa \(2026\)](#): an exact match on calendar year-month, sex, Census region, and a five-bin baseline-FPL income group, followed by propensity-score matching within cell on a vector $\mathbf{W}_i$ of baseline covariates (demographics, family structure, job characteristics, baseline insurance, and household income relative to FPL). Treated-control pairs are formed by greedy nearest-neighbor matching on the score $\hat{p}_i$, using the standard scalar-score-balances-vector result ([Rosenbaum and Rubin, 1983](#)). The final matched sample contains 1,889 treated and 1,820 control workers (95.3% match rate); within-pair propensity-score distance is tight (median 3×10^{-5}) and standardized mean differences fall below the 0.10 balance threshold ([Austin, 2011](#)) on every covariate. Full cell construction, the covariate vector $\mathbf{W}_i$, the logit specification, the greedy matching algorithm, and balance diagnostics are in [Appendix B](#). The components are labeled "Step 1" and "Step 2" to avoid collision with the two-part hurdle stages of [Section 4.5](#).

4.3 Reweighting for cross-era comparability

Matching produces well-balanced pairs within each era, but the composition of displaced workers differs across eras (the workers displaced in 2006 are not the workers displaced in 2016), so a pre/post difference in $\hat{\beta}_k$ could reflect compositional shifts rather than the policy. Following [Schaller and Zerpa \(2026\)](#), I reweight the pre-ACA matched pairs to resemble the post-ACA pairs on a vector $\mathbf{Z}_i$ of baseline demographics. I estimate a second logit of post-ACA-cohort membership on $\mathbf{Z}_i$, then assign each pre-ACA matched pair an inverse-odds pair weight that enters all downstream regressions. Critically, $\mathbf{Z}_i$ excludes baseline earnings and baseline insurance, because those are outcomes of the policy environment and reweighting on them would partially absorb the ACA effect. Balance checks confirm the demographic distributions are statistically indistinguishable after reweighting; a robustness specification additionally incorporating the MEPS person weights yields nearly identical estimates ([Table 9](#)). The full $\mathbf{Z}_i$ specification, the post-ACA logit, the inverse-odds weight formula, and the balance diagnostics are in [Appendix B](#).

4.4 Event-study specification

The matched, reweighted panel is a stacked event-time dataset in which each control inherits the event month of its matched treated worker. I define bimonthly event-time bins

$$k \in \{-6, -4, -2, 0, +2, +4, +6, +8, +10, +12\},$$

where each bin spans two calendar months relative to the displacement date. The reference is $k = -6$ (omitted bin, covering months -6 and -5 before separation); $k = 0$ covers the month of separation and the month immediately after; $k = +2$ covers months 2 and 3 after; and so on. This is the same bimonthly indexing used by [Schaller and Zerpa \(2026\)](#). Headline tables report the central window $k \in \{-4, -2, 0, +2, +4\}$, which spans roughly ten months around displacement and captures the peak coverage loss.

Estimating equation. To estimate how the displacement effect differs across policy environments, I interact event-time dummies with three mutually exclusive era indicators: Pre_i (baseline before 2014 excluding the Great Recession), GR_i (2008 to 2010), and Post_i (2014 and later). Letting $e \in \{\text{Pre}, \text{GR}, \text{Post}\}$ index the eras, the estimating equation is

$$Y_{it} = \mu_i + \alpha_t + \sum_{k \neq -6} \sum_e \left[\lambda_k^e D_{ikt} \mathbb{1}\{e_i = e\} + \beta_k^e D_{ikt} T_i \mathbb{1}\{e_i = e\} \right] + \varepsilon_{it}. \quad (1)$$

The dependent variable Y_{it} is one of the five outcome variables defined in Section 3.3. The fixed effects μ_i (person) and α_t (calendar month-year) absorb time-invariant individual heterogeneity and aggregate time trends. The event-time dummy is $D_{ikt} = \mathbb{1}\{t - E_i = k\}$, defined relative to worker i 's event month E_i ; $T_i \in \{0, 1\}$ is the treatment indicator (one for displaced workers, zero for matched controls); $e_i \in \{\text{Pre}, \text{GR}, \text{Post}\}$ is the era to which worker i 's baseline belongs; and ε_{it} is a person-clustered error term. The era-specific coefficients λ_k^e capture the bin-by-era event-time pattern in the matched-control sample, and β_k^e captures the additional bin-by-era event-time pattern attributable to having been displaced. The era-specific coefficients β_k^{Pre} and β_k^{Post} are the central estimands. Equation (1) mirrors equation (2) in [Schaller and Zerpa \(2026\)](#) and produces every event-study result in Section 5. I estimate it by weighted within transformation using the matched-pair weight and cluster standard errors at the person level. For binary outcomes I estimate equation (1) as a linear probability model so that coefficients are directly interpretable as changes in coverage probability in percentage points; the cross-era contrast then has the same interpretation without the latent-scale conversions a logit or probit would require.

Era-specific coefficients and the cross-era contrast. Pre-ACA and post-ACA workers form disjoint clusters in the matched panel (each worker is in precisely one era), so $\hat{\beta}_k^{\text{Pre}}$ and $\hat{\beta}_k^{\text{Post}}$ are estimated independently. The descriptive cross-era contrast is

$$\hat{\Delta}_k = \hat{\beta}_k^{\text{Post}} - \hat{\beta}_k^{\text{Pre}}, \quad k \in \{-4, -2, 0, +2, +4\}, \quad (2)$$

which I refer to as the per- k ACA mitigation. For coverage outcomes, $\hat{\Delta}_k > 0$ indicates that the post-ACA displacement effect is smaller in absolute magnitude than the pre-ACA effect. The standard error of the contrast is the independent-variance form $\sqrt{\widehat{\text{Var}}(\hat{\beta}_k^{\text{Post}}) + \widehat{\text{Var}}(\hat{\beta}_k^{\text{Pre}})}$ because the cross-era covariance is zero by construction (Appendix C). The contrast is mechanically noisier than either era-specific coefficient, so per- k contrasts can fail to clear conventional significance thresholds even when the era-specific coefficients are sharply identified. Schaller and Zerpa (2026) report paired pre- and post-ACA point estimates without testing the contrast directly; reporting $\hat{\Delta}_k$ and its standard error explicitly is a small refinement of that approach.

4.5 Two-part hurdle for the burden outcomes

The five outcome variables defined in Section 3.3 group into two analytical dimensions. The *coverage* dimension consists of the three person-month indicators (any insurance, private, public). Each is a valid Y_{it} for a stand-alone event-study estimation of equation (1), with no further decomposition needed. The *cost burden* dimension, consisting of the out-of-pocket premium and the burden ratio, requires more care. The burden ratio is undefined for household-months without private coverage and equals zero by construction for household-months in which the employer pays the entire premium. Estimating equation (1) on burden_{it} directly therefore averages over two substantively distinct populations: workers whose plan generates no premium contribution, and workers who pay positive premiums. Following Goldman et al. (2018), I decompose the burden dimension into a two-part hurdle that separates the *extensive* and *intensive* margins. Following standard usage, the *extensive margin* is whether premiums are paid at all (a 0/1 indicator, Stage 1A below), and the *intensive margin* is the burden share conditional on payment (Stage 2 below).

Stage 1A (extensive: probability of paying any premium). With $Z_{it} = \mathbb{1}\{\text{premium}_{it} > 0\}$ as the outcome, Stage 1A applies the form of equation (1) with era-specific coefficients renamed $\beta_k^{1A,e}$. Estimation is on the full matched panel of $N = 43,989$ person-months, which captures both the extensive coverage margin and the within-covered composition shift toward premium-bearing plans. Stage 1A answers the policy question: of all displaced workers, what fraction ends up paying any premium?

Stage 2 (intensive: burden among premium-payers). Stage 2 applies the form of equation (1) with era-specific coefficients $\beta_k^{2,e}$, with the burden ratio and the high-burden indicator $\mathbb{1}\{\text{burden}_{it} > 0.10\}$ as outcomes, on the within-payer subsample $\{(i, t) : \text{premium}_{it} > 0, Y_{it}^{\text{HH}} \geq \underline{Y}\}$ of $N = 14,097$ person-months on 1,663 workers (1,057 treated). Stage 2 isolates a within-payer price effect: among workers who pay premiums after displacement, by how much does displacement raise the burden share?

Stage-specific cross-era contrasts. The decomposition produces era-specific coefficients in each stage, with corresponding per- k ACA mitigation contrasts

$$\hat{\Delta}_k^{1A} = \hat{\beta}_k^{1A,\text{Post}} - \hat{\beta}_k^{1A,\text{Pre}}, \quad \hat{\Delta}_k^2 = \hat{\beta}_k^{2,\text{Post}} - \hat{\beta}_k^{2,\text{Pre}}, \quad (3)$$

each with standard errors computed as for $\hat{\Delta}_k$ in equation (2). As in the headline specification, the era-specific coefficients are the primary inferential objects. The full Stage 1A and Stage 2 estimating equations (equations (9) and (10)), the Stage 1B composition diagnostic (which re-estimates the Stage 1A premium-paying indicator on the privately-covered subsample only, isolating the within-covered composition shift), the single-stage reconciliation, the log-form Stage 2 robustness check, and the substantive departures from [Goldman et al. \(2018\)](#) are in Appendix B.

4.6 Heterogeneity: latent class analysis and a triple-difference test

The event study returns an *average* ACA mitigation. To ask whether that average masks heterogeneity across types of jobs, I follow [Brauner et al. \(2019\)](#) and cluster workers into latent classes on the basis of 23 baseline binary indicators measured at $k = -6$ (thirteen job-quality features, six industry dummies, four occupation-group dummies). The latent class model represents each worker as a draw from one of K classes:

$$\Pr(\mathbf{x}_i) = \sum_{k=1}^K \tau_k \prod_{j=1}^J \pi_{jk}^{x_{ij}} (1 - \pi_{jk})^{1-x_{ij}}, \quad J = 23, \quad (4)$$

where τ_k is the class share and π_{jk} is the probability that indicator j equals 1 in class k . Estimation is by EM with 25 random starts. Following [Brauner et al. \(2019\)](#)’s multi-criterion logic, I select $K = 4$ as the largest K at which information criteria, classification quality, minimum class size, and substantive interpretability are jointly satisfied. To facilitate the formal heterogeneity test, I aggregate the four classes into two super-classes (*Good-job* and *Precarious*) using a stable data-driven labeling rule based on baseline employer-coverage offer probability. Throughout the figures and tables I also use the abbreviations HQ for the Good-job super-class and LQ for the Precarious super-class; these reflect the latent “high-quality” / “low-quality” job-characteristic profile the

largest-gap rule recovers and are the variable names in equation (5). Indicator construction with explicit cutoffs, K -selection diagnostics, the labeling rule numerics, and the four-class profiles are in Appendix D.

Per-class and triple-difference event studies. The heterogeneity analysis re-estimates equation (1) separately within each class and within each super-class. For a formal test of whether ACA mitigation varies by job type, I estimate a triple-difference (DDD) specification on the full panel:

$$Y_{it} = \mu_i + \alpha_t + \sum_{k \neq -6} \sum_e \beta_k^e D_{ikt} T_i \mathbb{1}\{e_i = e\} + \sum_{k \neq -6} \sum_e \phi_k^e D_{ikt} T_i \mathbb{1}\{e_i = e\} \text{HQ}_i + \text{controls} + \varepsilon_{it}, \quad (5)$$

where HQ_i indicates Good-job super-class membership and “controls” absorbs the era main effects and the HQ main effect terms. The joint Wald test on $\{\phi_k^{\text{Post}} - \phi_k^{\text{Pre}}\}_k$ across nine event-time bins (9 d.f.) directly tests whether ACA mitigation differs between the Good-job and Precarious super-classes; per- k contrasts and the joint test appear in Table 7, Panels C and D.

5 Results

This section presents the four analyses set out in Section 4.1: the headline event study with two-part hurdle decomposition (Section 5.1); a baseline-ESI-exposure test of the operative mechanism (Section 5.2); a stratification by baseline income (Section 5.3); and a worker-type heterogeneity analysis using a latent-class partition (Section 5.4). Section 5.5 reports the design diagnostics that apply to all four.

5.1 Main event study: coverage and burden

I estimate equation (1) on the full matched panel separately by era and recover the two-part hurdle decomposition for the burden outcomes (Section 4.5). The headline result, reported in Table 3 and plotted in Figure 5, has two parts. First, post-ACA displaced workers lose meaningfully less coverage than their pre-ACA counterparts. Second, among workers who retain private coverage and pay premiums after displacement, the within-payer burden is no lower post-ACA and is meaningfully higher on the high-burden share. The policy works through coverage substitution, not through making the coverage that the displaced retain any cheaper.

Insurance coverage. In both eras, coverage is flat through the pre-treatment window and drops sharply at the bin covering the month of separation. The size of this immediate drop is comparable across eras: the ACA is not designed to prevent employer plans from ending when the job

ends, only to change what happens afterward. The post-ACA gap is 4.9pp smaller at two months and 6.1pp smaller at four months than the pre-ACA gap, an absorption of roughly one-quarter of the two-month coverage loss and one-third of the four-month loss; the gap operates through both partial private-margin substitution and a modest increase in public-coverage take-up (era-specific point estimates and significance in Table 3, panel 1). The era-specific coefficients are individually significant at 1% in both eras at every post-displacement horizon. The per-horizon mitigation contrast does not individually clear conventional thresholds because of the SE inflation that differencing two coefficients introduces (Appendix C).

Premium burden: the two-part decomposition matters. The two-part hurdle separates a coverage outcome (whether the worker pays any premium after displacement) from an affordability outcome (the burden share conditional on paying). The two stages move in opposite directions, and a single-stage burden specification that does not separate them produces a near-zero summary that hides both effects.

The *extensive margin* (Stage 1A) shows the ACA shifting workers *onto* premium-paying plans after displacement. The post-ACA probability of paying any premium falls less sharply than the pre-ACA probability, with a four-month cross-era contrast of +8.2pp (significant at 10%); era-specific estimates and the rest of the horizon profile are in Table 3, panel 4. The pattern is consistent with non-employer private plans absorbing a fraction of the workers who would otherwise have lost coverage entirely. A composition-isolation diagnostic restricted to the privately covered (Stage 1B) yields a sharper version of this result, with mitigation contrasts roughly twice as large; details in Appendix B.

The *intensive margin* (Stage 2) moves in the opposite direction on the policy-relevant high-burden indicator. The post-ACA displacement effect more than doubles the pre-ACA effect at two months on the share of premium-payers crossing the 10% household-income threshold, with a cross-era contrast of +15.0pp significant at 5% (Table 3, panel 6). The continuous burden ratio (panel 5) shows the same direction with a smaller, individually insignificant contrast. Among workers who pay premiums after displacement, the post-ACA distribution has shifted toward the high-burden right tail. This pattern aligns with population-level evidence in [Goldman et al. \(2018\)](#) that mean post-ACA premium contributions rose for non-subsidy-eligible households; the displacement-specific finding here extends that pattern to the moment a worker crosses into a vulnerable state.

Read jointly, the two stages give the headline a sharper interpretation: *coverage substitution with rising conditional burden*. The ACA shifted displaced workers onto premium-paying non-employer private plans, predominantly Marketplace plans, but the conditional burden on those plans rose. Marketplace coverage is real insurance, better than no coverage at all; it is not cheap

insurance for the displaced.

5.2 ESI exposure and post-displacement coverage loss

I estimate equation (1) on the pre-ACA matched panel separately within three groups defined by the displaced worker’s baseline ESI exposure at $k = -6$. Group A is workers covered through their own ESI; their coverage is tied directly to the job that ends. Group B is workers offered ESI but covered through a spouse’s plan; their own ESI is available but unused, so displacement does not sever their actual coverage source. Group C is workers not offered ESI at all. Restricting to the pre-ACA sample isolates the ESI tie at the moment it was binding. The test discriminates the ESI dependency from generic consequences of displacement (income loss, loss of any employer benefit) that would affect coverage through any channel: under the ESI-dependency reading, the loss should be concentrated in Group A and near-zero in Group C; under the generic-consequences reading, the three groups should show similar loss.

The estimates in Table 4 and Figure 6 support the ESI-dependency reading. Group A loses substantial coverage after displacement (a 42.3pp loss at the four-month horizon, significant at 1%); Group B loses much less (4.6pp loss, not statistically distinguishable from zero); and Group C shows essentially no displacement-induced coverage loss at all, since their coverage was not tied to the job in the first place. The monotone ordering across the three groups is what the [Madrian \(1994\)](#) mechanism predicts and is hard to generate under any reading that does not condition on baseline coverage. The result sharpens the headline by identifying the population whose post-displacement coverage shock the ACA can plausibly reach: workers whose baseline coverage was tied to their job.

5.3 Income heterogeneity

The income heterogeneity check follows [Schaller and Zerpa \(2026\)](#)’s middle-income finding directly: they document that post-ACA coverage gains concentrate in workers near the Medicaid-expansion threshold and the lower portion of the Marketplace-subsidy range, and the income stratification here tests whether the same pattern appears on MEPS at the displacement margin. I re-estimate the matched event-study specification of equation (1) separately within four baseline-income bands on the matched panel, and check whether the post-ACA mitigation appears in the income ranges the ACA’s subsidy and expansion thresholds were designed to reach. The four bands are Poor (below 100% FPL), Low Income (100–199% FPL, spanning the Medicaid-expansion threshold at 138% FPL and the lower portion of the Marketplace-subsidy range), Middle Income (200–399% FPL), and High Income (400% FPL and above); the band boundaries follow the standard MEPS poverty categories, with the merge rationale described below; the model and matched

sample are otherwise identical to Section 5.1. The headline result, reported in Table 8, is sharply concentrated. The post-ACA trajectory diverges from the pre-ACA trajectory in the Low Income band the policy thresholds reach, with no measurable mitigation in the Middle and High bands and only a small effect at separation in the Poor band. The pattern is what the policy architecture predicts, and the stratified evidence sharpens the headline.

The four-bin specification used here merges the small Near Poor cell (100–124% FPL, post-ACA $N = 56$) with the Low Income cell (125–199% FPL) into a single Low Income (100–199% FPL) band. The merge is undertaken to address a small-cell power problem in the Near Poor cell and to reduce the pre-trend sensitivity it introduces. The original five-bin classification produces a Tier 3 pre-trend rejection on the Low Income (125–199%) any-insurance mitigation contrast, consistent with previously-eligible-but-unenrolled workers picking up Medicaid coverage in this band after 2014, which is itself a pattern of substantive interest. The merged 4-bin specification passes the joint pre-trend test on the mitigation contrast cleanly ($\chi^2_2 = 0.51$, $p = 0.77$) and is the income-heterogeneity object reported below; the original 5-bin estimates are available on request. The merge is local to this stratification; the matching cells retain the original five categories.

The Low Income band is where the policy bites. Pre-ACA Low Income workers lose roughly fifteen percentage points of any insurance at two and four months post-displacement, the canonical coverage shock. Post-ACA Low Income workers show essentially flat trajectories, with the cross-era contrast reaching +17.4pp at four months (significant at 10%; see Table 8). The merged band pools three sub-ranges the ACA treats differently (100–138% Medicaid-expansion eligibility in expansion states, 125–138% Medicaid and subsidies overlap, and 138–199% Marketplace subsidies operative), and the result reads as the combined policy effect across all three channels.

The other three bands fit the policy architecture. The Poor band (below 100% FPL) shows a positive +12.9pp mitigation at separation (significant at 10%) with diffuse estimates at later horizons, consistent with Medicaid-expansion coverage being available irrespective of displacement timing. The Middle Income band (200–399% FPL) and High Income band ($\geq 400\%$ FPL) are substantive non-findings: the pre-ACA and post-ACA event-time trajectories sit on top of one another, and the cross-era difference is indistinguishable from zero at every horizon. This is what the policy architecture predicts: Middle Income workers are mostly above the Medicaid expansion ceiling and in the upper range of subsidy eligibility where the schedule provides smaller absolute contributions, while High Income workers were ineligible for Marketplace subsidies and uninvolved in the Medicaid expansion. Across bands, the post-ACA trajectory diverges from the pre-ACA trajectory exactly where the policy thresholds predict, and the stratified pattern is the sharpest test the data offer that the post-2014 provisions operated as intended.

5.4 Worker-type heterogeneity: latent classes and the per-class event studies

In this subsection, I ask whether the headline mitigation pattern varies across types of workers, where “type” is defined empirically by clustering on baseline job characteristics rather than by income alone. Income is one heterogeneity axis, but it is not the only one. A displaced retail worker and a displaced software engineer at the same household income face different post-displacement coverage options because the underlying job characteristics differ in ways the income variable alone does not capture. The analysis proceeds in three stages. The latent-class model of equation (4) first partitions all matched workers into $K = 4$ classes on the empirical distribution of 23 baseline job-quality indicators, with the four classes aggregated into Good-job and Precarious super-classes by the data-driven labeling rule. The matched event-study specification of equation (1) is then re-estimated separately within each of the four classes on the panel restricted to that class. Finally, the triple-difference specification of equation (5) interacts the era-specific event-time coefficients with super-class membership to deliver a formal joint test of whether ACA mitigation differs between Good-job and Precarious workers.

5.4.1 Class structure

The classification recovers four classes whose substantive identities map cleanly onto familiar labor-market contours. Class 3 has near-universal full-time work, the highest baseline employer coverage offer in the sample, and the highest non-wage benefit coverage: the “Good-job Finance and Professional” class. The other three classes share much lower employer coverage offer rates and much weaker benefit coverage, but differ on occupation and industry: Class 1 (Service), Class 2 (Trade and Sales) with high temporary-job exposure, and Class 4 (Goods and Blue-collar) concentrated in manufacturing and construction. The data-adaptive largest-gap rule places Class 3 alone above the gap (the Good-job super-class) and the three precarious classes below it (the Precarious super-class). Class profiles appear in Figure 7; share statistics in Table 5.

The four classes are not isomorphic to income bins: Class 3 spans middle and high-income bands, and the three precarious classes overlap on income but differ on occupation, temporary-job exposure, and benefit structure. The classification captures *labor-market type* independent of the income axis in Section 5.3.

5.4.2 Per-class event study

Re-estimating equation (1) within each of the four classes produces a clean three-way split, reported in Table 6 and Figures 8–10. Pre-ACA displacement effects scale with baseline employer coverage exposure as expected: the Good-job class (Class 3) loses the most because it had the most to lose, and the three precarious classes lose substantially less. The mitigation pattern is the more

substantively interesting result.

Good-job class (C3). C3 shows the cleanest mitigation in the paper: +16.4pp on any insurance at $k = +4$ (significant at 1%) and +11.8pp on private coverage at $k = +4$ (significant at 10%), with the joint pre-trend test passing. The pattern is consistent with Marketplace substitution: workers whose income remains within subsidy range after displacement substitute Marketplace coverage for the lost employer plan.

Service class (C1). C1 shows positive but imprecisely identified mitigation point estimates. One private-coverage contrast at $k = +4$ is individually significant at 10% (+9.8pp); the others are not. The qualitative direction is consistent with C3, but the precision is too low to draw a sharp conclusion.

Trade and Sales class (C2). C2 shows a negative any-insurance contrast at $k = +4$ (−20.2pp, $p < 0.10$) that a naive reading would attribute to the ACA worsening coverage. This reading is not supported by the pre-displacement leads: the post-ACA C2 cohort’s coverage trajectory is already downward-sloping before separation, with cohort estimates at $k = -4$ and $k = -2$ both individually significant at 5% (Table 6). The post-displacement coefficient is more parsimoniously read as a cohort-level coverage decline in the post-2014 period, not a displacement-specific ACA effect. I retain the row as descriptive evidence of the cohort pattern and do not interpret the $k = +4$ point estimate causally.

Goods and Blue-collar class (C4). C4 shows essentially zero mitigation across all outcomes and post-displacement horizons (Table 6); no detectable ACA effect.

Reading the per-class pattern. Eligibility alone does not explain the per-class split. Workers in the three precarious classes are formally eligible for the Marketplace and Medicaid on the same income thresholds as everyone else, yet their post-2014 coverage trajectories show no measurable mitigation. The data here cannot distinguish whether this gap reflects an eligibility-utilization shortfall, a take-up infrastructure deficit, or both. One reading consistent with the pattern is that take-up rather than eligibility is the binding margin: enrolling in Marketplace coverage and keeping it active requires resources that are unevenly distributed across industries and worker types. Two such resources are help with the enrollment process itself, which Marketplace navigators (federally-funded counselors created under the ACA) and similar state and community-based programs provide, but at very different intensities across regions and industries; and stable enough month-to-month income to keep up with monthly premium payments, which is harder for workers

in industries with episodic or intermittent post-displacement employment. The data here cannot test these channels directly. Related evidence in [Aboulafia and Schneider \(2025\)](#) on the ACA’s employer-mandate provision documents that the same precarious-worker population is systematically excluded from the employer-coverage channel by firm-size exemptions, part-time hour thresholds, and franchise structures. While that paper studies a different ACA provision than this one, the two findings point in the same direction: the ACA’s various coverage channels reach precarious workers unevenly, and extending eligibility on paper does not translate equally into coverage for this population.

The per-class burden estimates are not reported in Table 6 because the Stage 2 sample requires both private coverage and a positive out-of-pocket premium, which leaves three of the four classes with post-ACA samples too small for stable estimation. The per-class affordability story is reported at the super-class level in the next subsection.

5.4.3 *Super-class joint test*

I estimate the triple-difference specification of equation (5) on the full matched panel, interacting the era-specific event-time coefficients with super-class membership (Good-job class C3 alone vs. the three Precarious classes pooled). The joint Wald test on the cross-era difference of these interaction terms across the nine event-time bins is the formal test of whether ACA mitigation differs between the two super-classes, conducted separately for each of six outcomes (the three coverage indicators, the Stage 1A premium-paying indicator, the Stage 2 burden ratio, and the Stage 2 high-burden indicator). Results are in Table 7, Panel D. The headline pattern has two parts: the test rejects super-class homogeneity for two extensive-margin outcomes (private coverage and the Stage 1A premium-paying indicator) and does not reject for the four other outcomes. The rejections concentrate the ACA’s premium-paying coverage gains in the Good-job super-class; the non-rejections show that public-coverage take-up and within-payer affordability moved similarly across worker types.

Extensive-margin rejections. Homogeneity is rejected on private coverage ($\chi^2_9 = 18.13$, $p = 0.034$) and on the Stage 1A premium-paying indicator ($\chi^2_9 = 18.39$, $p = 0.031$). Both findings are on the extensive coverage margin, and they tell a consistent story: the post-ACA shift toward premium-bearing private coverage after displacement was concentrated in the Good-job super-class. Workers in the precarious classes did not experience the same post-2014 substitution from lost employer coverage onto premium-paying Marketplace plans, which matches the per-class results in Section 5.4.2 and reinforces the take-up reading there.

Non-rejections. The non-rejection on public coverage is consistent with Medicaid eligibility being income-determined rather than job-type-determined: once income is held constant, Medicaid take-up after displacement moves similarly across worker types. The non-rejections on the two Stage 2 burden outcomes (the burden ratio and the high-burden indicator) say that among workers who pay premiums after displacement, the post-ACA distribution shifted toward the high-burden tail by similar magnitudes in both super-classes, even though the precision of the LQ within-payer estimate is limited.

The within-LQ Stage 2 sample is small and selected by construction: it captures only the precarious workers who actually land on premium-paying private plans after displacement, and most precarious workers in the data either remain uninsured or migrate to Medicaid (which removes them from the Stage 2 sample). The larger LQ affordability story therefore operates through Medicaid take-up rather than through the Stage 2 high-burden margin, which the Stage 1A DDD rejection and the public-coverage event study at the per-class level both pick up. Among the LQ workers who do land on Marketplace coverage, the within-payer burden pattern resembles HQ workers', which fits the policy architecture: Medicaid expansion targets LQ income bands, Marketplace subsidies target middle-income bands, and within the small group of LQ workers who use the Marketplace, the cost pattern is the same as the larger HQ Marketplace population's.

Synthesis. The job-quality classification reveals patterns the income axis does not. ACA coverage mitigation accrues to the Good-job professional class strongly, to the Service class weakly, to the Trade and Sales class not at all (and the negative point estimate there is read as cohort-level decline, not an ACA effect), and to the Goods and Blue-collar class not at all. This pattern parallels the income, race, age, and marital-status heterogeneity [Courtemanche et al. \(2019\)](#) documents at the population level, and extends the heterogeneity story from those demographic axes to a job-type axis the income variable alone does not capture.

5.5 Design diagnostics and paper-wide robustness

In this subsection, I summarize the design diagnostics that apply to every result in Sections 5.1–5.4. The matched event-study design rests on three identifying assumptions: parallel pre-trends in the outcomes, comparable treatment intensity across eras, and a real structural break at the ACA's 2014 implementation rather than a smooth secular trend. All three are supported.

Joint Wald tests of zero pre-displacement leads are non-rejecting on the policy-relevant mitigation contrast for every specification in the paper. Across 59 estimated event-study specifications spanning the main coverage trio, the two-part hurdle, the coverage theory test, income heterogeneity, the per-class and super-class LCA stratifications, and the HQ-vs-LQ DDD contrasts, the joint test $H_0 : \beta_{-4}^{\text{Post}} - \beta_{-4}^{\text{Pre}} = \beta_{-2}^{\text{Post}} - \beta_{-2}^{\text{Pre}} = 0$ fails to reject at the 5% level in every case (Appendix C).

Three specifications are near the 10% level: two below, the per-class C2 Trade and Sales any-insurance contrast at $p = 0.082$ and the super-class Precarious Stage 2 burden-ratio contrast at $p = 0.092$, and one just above, the per-class C2 public-coverage contrast at $p = 0.103$. These are disclosed in the relevant results subsections.

The employment shock at separation is nearly identical across eras, ruling out a confound from differential displacement severity. Panel-by-panel estimates of the peak coverage shock cluster tightly above twenty percentage points in panels covering pre-2014 years and drop sharply in panels covering 2015–2018, consistent with a 2014 break and inconsistent with a smooth secular trend (Appendix Figure 21). Table 9 reports five alternative specifications, including dropping the Great Recession years, classifying them as part of the pre-ACA period, and dropping reweighting altogether. The headline mitigation point estimate is stable across all five.

6 Discussion

This section addresses four issues that bound the interpretation of the findings: what the design identifies and does not identify, the limitations of scope, the mechanism question, and the policy implications.

Identification. The matched event study identifies the causal effect of *involuntary displacement* on insurance coverage and on the within-payer premium burden, separately by policy era. It does not identify pre-separation *job lock* in the original sense, where workers stay in jobs they would otherwise leave because leaving means losing coverage (Madrian, 1994; Gruber and Madrian, 2002). That question would require variation in whether a worker has employer coverage at all, which the displacement event does not provide. What the design provides is reduced-form evidence on one observable consequence of the ESI tie that all job-lock theories share, namely that ending the job ends coverage for workers whose coverage came through that job, and a measurement of the extent to which the post-2014 ACA architecture softens the resulting coverage shock. The size of the post-displacement coverage loss across the three baseline ESI groups in Section 5.2 is the qualitative test of the ESI mechanism; the era-specific event-study coefficients quantify the mitigation. The design does not separately identify which channel within the ESI tie drives the post-displacement coverage loss: it could be affordability after income loss, switching costs and inattention in moving to non-employer plans, information frictions in navigating new programs, or some combination. Sorting these out would require additional variation the matched-displacement design cannot supply. The patterns documented here are best read as a set of empirical regularities a structural model could try to reproduce; Aizawa and Fang (2020) and Fang and Shephard (2019) provide the structural welfare counterparts, and the reduced-form evidence here comple-

ments rather than substitutes for that work.

Limitations. Several features of the analysis limit how far the findings travel. First, the sample period ends in 2018, before the individual-mandate penalty was set to zero (effective 2019), the expansion of short-term limited-duration plans (2018–2019 federal rule), and the American Rescue Plan (2021) and Inflation Reduction Act (2022) subsidy enhancements. Whether the post-displacement mitigation documented for 2014–2018 has held up through these later policy changes is an open question. Second, the matched sample contains roughly 1,900 treated workers. This is enough to estimate each era’s event-study coefficients precisely (most are significant at 1%) but not enough to estimate the per-horizon cross-era difference between them precisely, because differencing two estimates inflates the standard error. Per-horizon contrasts can therefore fail to clear conventional thresholds even when both era-specific coefficients are sharply identified, and the income-stratified and per-class cells, which are smaller still, are noisier on this count.

Two further limitations follow from the structure of the MEPS public-use files, and a third concerns the take-up channel. First, MEPS public-use provides a continuous family-income-to-FPL ratio variable (POVLEV) only beginning in 2007. For sample years 2004–2006, only the five-category baseline poverty classification (POVCAT) is available. This constrained the income-heterogeneity stratification in Section 5.3 to the original POVCAT bins, with the merge of the smaller Near Poor cell into the Low Income cell carried out at the bin level rather than at a specific FPL number. Second, MEPS public-use does not include state identifiers, so the treatment is operationalized as the national 2014 shock and state-level variation in Medicaid expansion timing is not separately identified. [Schaller and Zerpa \(2026\)](#) show on SIPP data that expansion-state and non-expansion-state coverage trajectories diverge after 2014, with most of the coverage smoothing concentrated in expansion states; that state-level variation is averaged out in the national specification used here. A state-by-expansion design on these data would require the AHRQ Data Center restricted-use file. Third, the practical infrastructure that helps workers actually enroll (Marketplace navigators, employer-based outreach, online versus in-person enrollment) is not directly observed in the data; its role is inferred from the worker-type heterogeneity in Section 5.4 rather than measured.

Mechanism. This paper documents *where* the coverage loss is concentrated: in workers whose baseline coverage came through their own job, in lower-income workers whose income falls into the bands the ACA’s subsidies and Medicaid expansion target, and in worker types whose post-displacement income and labor-market position make Marketplace enrollment easier to sustain. It does not document *why* workers respond as they do. [Dey and Flinn \(2005\)](#) make this point sharply for the mobility version of the question: in their search-and-bargaining model, workers in insured

jobs stay longer because the insurance is productively valuable to the match, not because they are trapped, so the reduced-form mobility differential is consistent with no welfare loss at all. The same caution applies here. The post-displacement coverage facts are statistical descriptions of what happens at displacement; reading welfare into them requires a structural model that takes a stand on how the ESI–wage bundle is set, on what workers are willing to pay for coverage continuity, and on the frictions in the pre-ACA non-group market (Aizawa and Fang, 2020; Fang and Shephard, 2019).

Behavioral channels in the health-insurance-choice literature could also help explain the post-displacement coverage gap. Domurat et al. (2021) run a field experiment on the Covered California Marketplace and find that simply being inattentive to enrollment deadlines or misperceiving subsidy availability depresses take-up, with these frictions concentrated among low-income consumers, the same group where this paper finds the largest ACA mitigation. Handel (2013) shows that workers stick with their existing health plan even when switching would clearly help them. Cardon (2020) documents that loss aversion contributes to plan-switching inertia of the kind Handel (2013) observes, with workers treating any plan other than their current one as a loss the gain rarely justifies. The matched-displacement design here cannot tell which of these channels is at work; what it provides is a set of empirical patterns a future structural model could try to reproduce.

Policy implications. The findings bear on ongoing policy debates in three ways. First, the income-stratified pattern is direct evidence that the post-2014 ACA architecture works as a partial decoupling of insurance from employment, with the largest effects in the income bands the subsidy and expansion thresholds were designed to reach. Extending these provisions, by adopting Medicaid expansion in the remaining non-expansion states or by raising Marketplace subsidy generosity, would broaden the affordability of coverage after job loss along the same income gradient.

Second, the worker-type heterogeneity identifies populations the policy reaches unevenly. Within the precarious super-class, the coverage gains where they appear come through the public-insurance channel rather than through Marketplace substitution, which suggests Medicaid is the operative channel for this group. Changes in Medicaid eligibility design would therefore matter more for precarious workers than for Good-job workers, whose coverage gains come through the Marketplace.

Third, the two-part hurdle decomposition implies that coverage gains and conditional cost burden should be tracked as two separate policy outcomes. The post-2014 architecture reduces post-displacement coverage loss in the income bands the policy reaches, but does not reduce the cost burden faced by the workers who do take up the new coverage. A combined measure averaging the two would obscure this offsetting movement, which is precisely what the single-stage burden specification this paper documents does.

7 Conclusion

This paper estimates the effect of the Affordable Care Act on the insurance and out-of-pocket cost trajectories of involuntarily displaced workers, using a matched event-study design adapted from [Schaller and Zerpa \(2026\)](#).

The headline finding is that post-ACA workers lose meaningfully less insurance coverage after involuntary displacement than their pre-ACA counterparts, and the gains are concentrated where the policy architecture predicted. The pooled mitigation contrast does not individually clear conventional significance thresholds, but the income-stratified estimates show clean positive mitigation in the merged Low Income (100–199% FPL) bin and at the separation horizon in the Poor band, with no measurable mitigation in the Middle and High Income bands. The pooled estimate averages a real policy effect across income bands the policy was not designed to reach.

The coverage gains do not come cheaply. A two-part hurdle decomposition recovers a pattern that a single-stage burden estimator obscures: displaced workers shift onto premium-paying non-employer plans (positive ACA mitigation on the Stage 1A premium-paying margin), while among workers who pay premiums after displacement, the post-ACA distribution shifts meaningfully toward the high-burden tail (Stage 2). Coverage gains and conditional cost burden need to be tracked separately: the post-ACA architecture closes the coverage gap for the displaced workers it reaches, but does not lower the price they pay to remain insured.

The benefits also accrue unevenly across worker types. The latent-class analysis identifies four classes, and the Good-job Finance and Professional class delivers the single largest mitigation in the paper. The three precarious classes show no coverage gains, and the formal super-class joint test rejects homogeneity on both extensive-margin findings (private coverage and the Stage 1A premium-paying margin). The most consistent reading is that take-up rather than eligibility is the binding margin: Marketplace and Medicaid eligibility are income-determined, but enrollment relies on industry- and occupation-specific resources that do not reach all worker types equally.

Read jointly, the three findings give a more textured picture of the ACA’s reach than any single pooled estimate can. The post-2014 architecture reduces post-displacement coverage loss in the income bands the policy thresholds target and in the worker types whose post-displacement income and labor-market position make sustained Marketplace participation feasible. It leaves a meaningful cost burden on the workers it does cover, and it reaches the precarious-work population mainly through public insurance rather than through Marketplace substitution. None of these findings are visible in a pooled coverage estimate alone; recovering them required the income, hurdle, and worker-type stratifications this paper develops. Whether the within-payer cost burden and the precarious-worker take-up gap narrow under the post-2018 ACA enhancements, or under further extensions of the architecture, is a question this design does not answer but is one the same MEPS

framework, applied to later panels and to data with state-level information, can address.

8 Tables and Figures

Table 1: Descriptive statistics: full working-age MEPS panel, 2004–2018

Variable	Pre-ACA	Post-ACA	Overall
PANEL A. Demographics (person-year)			
Age	33.9 (22.5)	36.1 (23.1)	34.6 (22.7)
<i>Sex</i>			
Male	47.7%	47.8%	47.8%
Female	52.3%	52.2%	52.2%
<i>Race/ethnicity</i>			
Hispanic	28.8%	30.1%	29.2%
Non-Hispanic White	43.9%	41.3%	43.0%
Non-Hispanic Black	18.8%	18.3%	18.7%
Asian	5.7%	6.6%	6.0%
Other	2.7%	3.8%	3.1%
<i>Education</i>			
Less than HS	38.4%	35.3%	37.4%
HS graduate	34.2%	31.9%	33.4%
Some college	10.5%	13.0%	11.3%
College grad	11.3%	12.6%	11.7%
Graduate	5.6%	7.2%	6.1%
<i>Marital status</i>			
Married	36.5%	35.2%	36.1%
Previously married	14.6%	15.6%	14.9%
Never married	48.9%	49.2%	49.0%
<i>Poverty category</i>			
Poor (<100% FPL)	22.1%	21.3%	21.8%
Near Poor (100-124% FPL)	6.7%	6.0%	6.4%
Low Income (125-199% FPL)	17.2%	16.2%	16.9%
Middle Income (200-399% FPL)	28.3%	28.2%	28.3%
High Income (>=400% FPL)	25.8%	28.3%	26.6%
<i>Region</i>			
Northeast	15.1%	15.3%	15.2%
Midwest	19.4%	19.4%	19.4%
South	38.2%	38.4%	38.3%
West	27.3%	26.9%	27.1%

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(Table 1, continued)

Variable	Pre-ACA	Post-ACA	Overall
Has children (age <18 in HIEU)	54.8%	50.0%	53.3%
HIEU income, annual (2016\$)	53,942 (55,460)	55,682 (59,430)	54,508 (56,787)
HIEU size	2.8 (1.6)	2.7 (1.6)	2.8 (1.6)
Poverty pct (% FPL, 2007+)	301.4 (287.2)	325.6 (314.5)	311.2 (298.8)
PANEL B. Job characteristics (person-month)			
Employed	59.3%	59.2%	59.2%
Full-time (>=35 hrs/wk)	57.3%	55.0%	56.5%
Multiple jobs	10.0%	10.0%	10.0%
Temporary job	4.8%	5.5%	5.0%
Seasonal job	3.7%	4.2%	3.9%
Employer offers ESI	53.1%	51.7%	52.6%
Paid sick or medical leave	21.0%	20.6%	20.9%
Paid vacation	33.0%	29.7%	31.9%
Pension	23.4%	22.5%	23.1%
Union member	9.3%	8.4%	9.0%
Large employer (50+ FTE)	17.9%	17.2%	17.6%
Hours worked / month	76.4 (96.0)	78.0 (96.5)	76.9 (93.9)
Base wage / month (2016\$)	1,177 (2,573)	1,290 (2,871)	1,214 (2,674)
Wage supplements / month (2016\$)	47.0 (508.3)	60.8 (656.5)	51.5 (561.0)
Total wage income / month (2016\$)	1,224 (2,711)	1,351 (3,092)	1,265 (2,841)
<i>Industry distribution (employed only, raw MEPS 11-cat)</i>			
Agriculture	1.7%	1.6%	1.7%
Construction	7.6%	6.5%	7.2%
Manufacturing	11.4%	10.1%	11.0%
Trade	14.1%	12.4%	13.5%
Transport/util	5.2%	5.0%	5.1%
Finance/prof	19.4%	18.4%	19.1%
Educ/health	22.5%	21.3%	22.1%
Leisure/hosp	10.1%	10.2%	10.2%
Other svcs	5.6%	5.0%	5.4%
Public admin	5.1%	4.7%	5.0%
Military	0.6%	0.4%	0.6%
<i>Occupation distribution (employed only, raw MEPS 8-cat)</i>			
Mgmt/prof	32.1%	31.7%	32.0%

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(Table 1, continued)

Variable	Pre-ACA	Post-ACA	Overall
Service	20.7%	20.2%	20.6%
Sales	10.2%	8.9%	9.8%
Office/admin	13.2%	11.2%	12.5%
Constr/extract	9.7%	7.8%	9.0%
Prod/transport	14.6%	13.4%	14.2%
Farm/fish	1.1%	0.9%	1.1%
Military	0.6%	0.5%	0.6%
PANEL C. Insurance coverage outcomes (person-month)			
Any insurance	78.2%	85.2%	80.5%
Any private coverage	49.2%	49.8%	49.4%
Any public coverage	35.3%	43.3%	37.9%
of which: ESI (employer/union private)	44.0%	42.0%	43.4%
PANEL D. Out-of-pocket premium and burden outcomes (person-month; valid-burden subsample)			
Monthly OOP premium (\$2016)	213.9 (266.5)	245.1 (289.4)	224.2 (274.7)
Has positive OOP	38.4%	39.9%	38.9%
Premium burden (valid sample)	6.7% (12.9)	6.9% (12.5)	6.7% (12.8)
High burden (>10%, valid sample)	16.7%	18.0%	17.1%
Valid-burden share of person-months	45.5%	45.7%	45.6%
Person-months	3,981,030	1,925,846	5,906,876
Person-years	347,368	167,274	514,642
Unique persons	199,708	118,366	301,111

Notes. Descriptive statistics on the full working-age MEPS panel (adults aged 26–59) covering 2004–2018, split by policy era (pre-ACA: baseline before 2014 excluding Great Recession; post-ACA: 2014–2018). Panel A reports demographics at the person-year level; Panels B–D report employment, insurance, and out-of-pocket characteristics at the person-month level. Industry, occupation, and employer type in Panel B are conditional on being employed. Categorical variables shown as percentages; continuous variables shown as mean (standard deviation in parentheses).

Table 2: Descriptive statistics: matched event-study sample at baseline

Variable	Pre-ACA	Post-ACA	Overall
PANEL A. Demographics (at baseline, k=-6)			
Age	39.9 (9.6)	40.6 (9.9)	40.0 (9.6)
<i>Sex</i>			
Male	52.2%	48.5%	51.6%
Female	47.8%	51.5%	48.4%
<i>Race/ethnicity</i>			
Hispanic	28.9%	27.5%	28.6%
Non-Hispanic White	41.6%	36.2%	40.7%
Non-Hispanic Black	21.7%	24.6%	22.2%
Asian	5.9%	7.3%	6.1%
Other	2.0%	4.4%	2.4%
<i>Education</i>			
Less than HS	20.6%	14.6%	19.6%
HS graduate	48.9%	42.0%	47.7%
Some college	13.3%	20.7%	14.6%
College grad	12.4%	15.8%	13.0%
Graduate	4.8%	7.0%	5.1%
<i>Marital status</i>			
Married	51.7%	46.8%	50.8%
Previously married	19.9%	17.2%	19.4%
Never married	28.5%	36.1%	29.7%
<i>Poverty category</i>			
Poor (<100% FPL)	15.4%	19.4%	16.1%
Near Poor (100-124% FPL)	6.3%	5.8%	6.2%
Low Income (125-199% FPL)	21.0%	14.7%	19.9%
Middle Income (200-399% FPL)	34.5%	34.0%	34.4%
High Income (>=400% FPL)	22.9%	26.1%	23.4%
<i>Region</i>			
Northeast	14.1%	12.0%	13.8%
Midwest	17.4%	13.6%	16.7%
South	39.8%	45.3%	40.7%
West	28.7%	29.1%	28.7%
Has children (age <18 in HIEU)	49.7%	45.6%	49.0%
HIEU income, annual (\$2016)	50,215 (43,470)	51,138 (48,400)	50,369 (44,324)

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(Table 2, continued)

Variable	Pre-ACA	Post-ACA	Overall
HIEU size	2.6 (1.5)	2.5 (1.6)	2.6 (1.5)
Poverty pct (% FPL, 2007+)	290 (235)	306 (246)	293 (237)
PANEL B. Job characteristics (at baseline)			
Full-time (≥ 35 hrs/wk)	80.7%	78.3%	80.3%
Multiple jobs	11.7%	12.0%	11.7%
Temporary job	9.3%	11.5%	9.7%
Seasonal job	6.4%	6.3%	6.4%
Employer offers ESI	69.4%	67.4%	69.0%
Paid sick or medical leave	46.8%	47.7%	47.0%
Paid vacation	58.1%	53.7%	57.3%
Pension	35.7%	36.4%	35.8%
Union member	8.1%	7.9%	8.1%
Large employer (50+ FTE)	40.4%	39.3%	40.2%
Hours worked / month	173 (57)	170 (60)	172 (57)
Base wage / month (2016\$)	2,169 (1,649)	2,067 (1,643)	2,152 (1,648)
Wage supplements / month (2016\$)	120 (739)	130 (725)	121 (736)
Total wage income / month (2016\$)	2,249 (1,681)	2,146 (1,680)	2,232 (1,681)
PANEL C. Insurance coverage outcomes (at baseline)			
Any insurance	66.2%	75.4%	67.7%
Any private coverage	55.8%	59.7%	56.5%
Any public coverage	11.5%	19.3%	12.8%
PANEL D. Out-of-pocket premium and burden outcomes (at baseline; valid-burden subsample)			
Monthly OOP premium (\$2016)	155 (162)	179 (182)	159 (166)
Treated (involuntarily displaced)	1,577	312	1,889
Controls (matched)	1,514	306	1,820
Total matched persons	3,091	618	3,709

Notes. Descriptive statistics on the matched event-study analytical sample at baseline ($k = -6$), split by policy era. Categorical variables are shown as proportions (%); continuous variables are shown as mean (standard deviation in parentheses). The matched sample contains 1,889 treated (involuntarily displaced) workers and 1,820 matched controls drawn through the two-stage matching procedure described in Section 4.2: exact match on calendar year-month $\times$ sex $\times$ Census region $\times$ income group, then greedy 1:1 propensity-score matching within cell. Compared with the full-panel descriptives in Table 1, the matched sample is older, more likely to be full-time, and has higher employer coverage offer rates, reflecting the tenure and not-self-employed selection.

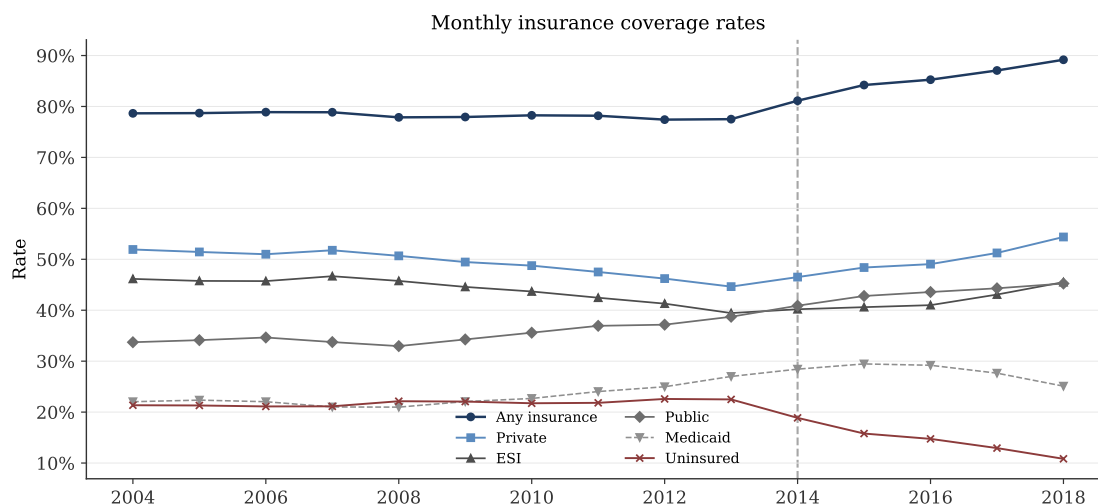


Figure 1: Monthly insurance coverage rates on the working-age MEPS panel, 2004–2018

Monthly insurance coverage rates on the working-age MEPS panel (adults 26–59), by calendar year. The figure plots six coverage indicators that overlap by construction (a person can hold both private and public coverage in the same month): any insurance, any private, ESI, any public, Medicaid, and uninsured. ESI is the dominant private channel throughout the sample window. Three patterns deserve note. First, ESI declines secularly pre-2014, from roughly 46% in 2004 to 39% in 2013. Second, after the 2014 ACA implementation marker, the uninsured share drops sharply (from roughly 23% to 11% by 2018), and both any-private (reflecting both Marketplace expansion and a modest ESI rebound back to 46%) and any-public coverage (reflecting Medicaid expansion, which grows from roughly 22% to 30% by 2015) rise. Third, the post-2014 ESI rebound runs against pre-ACA crowd-out concerns and indicates that the ACA’s non-employer coverage channels operate alongside employer coverage rather than displacing it.

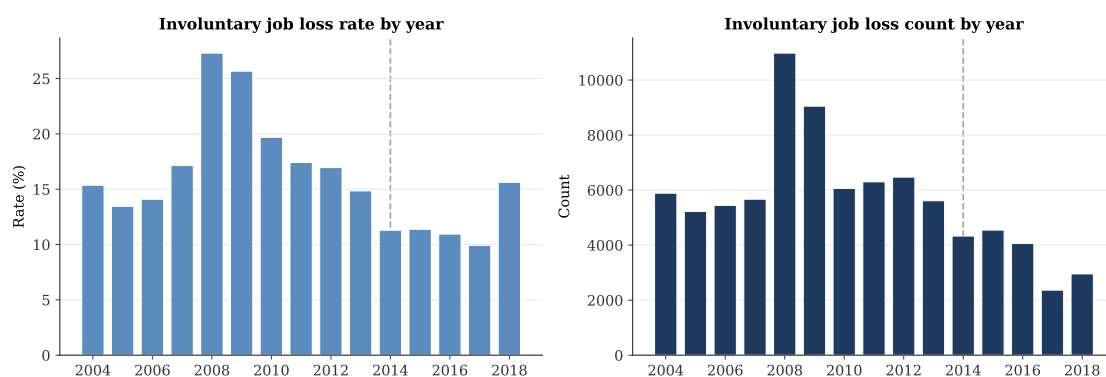


Figure 2: Incidence of involuntary job loss on the working-age panel, 2004–2018

Annual rate (left subplot) and count (right subplot) of person-months flagged as involuntary job loss (layoff, firing, or business closure) in the working-age MEPS sample. The spike during the 2008–2010 Great Recession and the subsequent decline motivate the Great Recession era control in Equation (1).

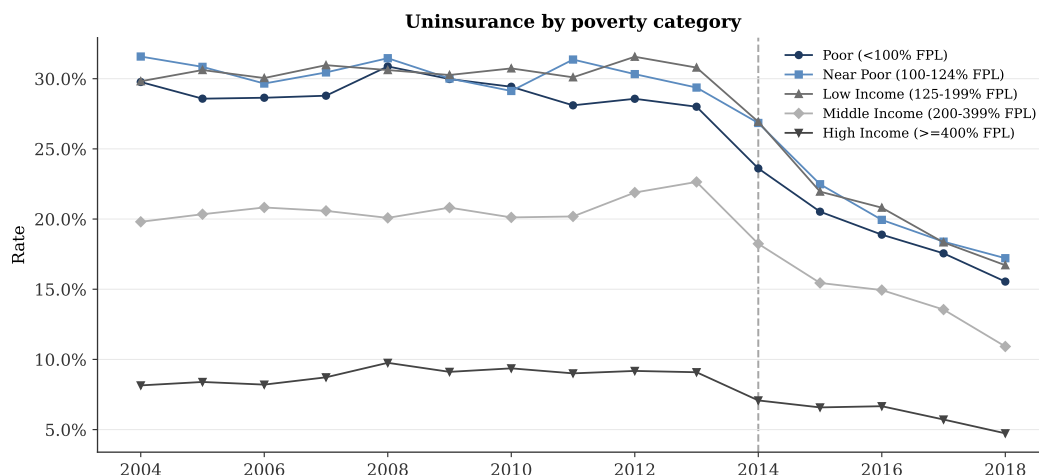


Figure 3: Uninsurance rate by baseline poverty status, 2004–2018

Monthly share of the working-age sample without any insurance coverage, disaggregated by five baseline poverty categories. The 2014 ACA shading marks the Marketplace launch and Medicaid expansion. The sharpest decline occurs among low-income workers, consistent with Medicaid expansion reaching the previously uninsured poor.

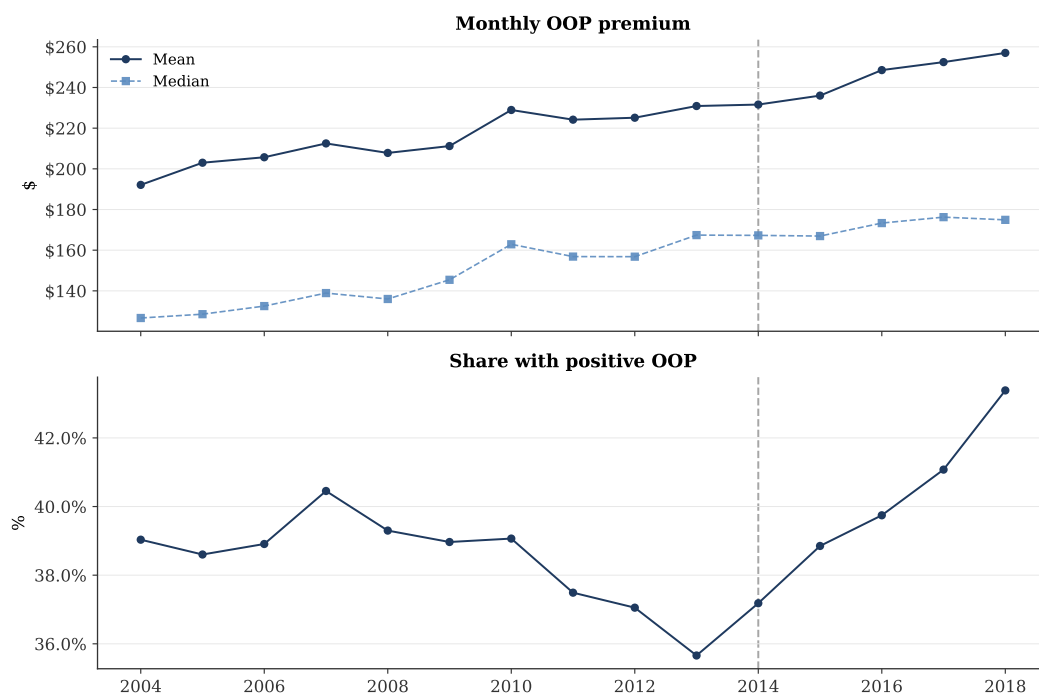


Figure 4: Out-of-pocket premium trends on the full working-age MEPS panel, 2004–2018

Top: mean and median monthly out-of-pocket premium contribution among person-months with positive premium, deflated to constant 2016 USD. Bottom: share of person-months with positive premium among privately-insured workers. The 2014 ACA implementation is marked.

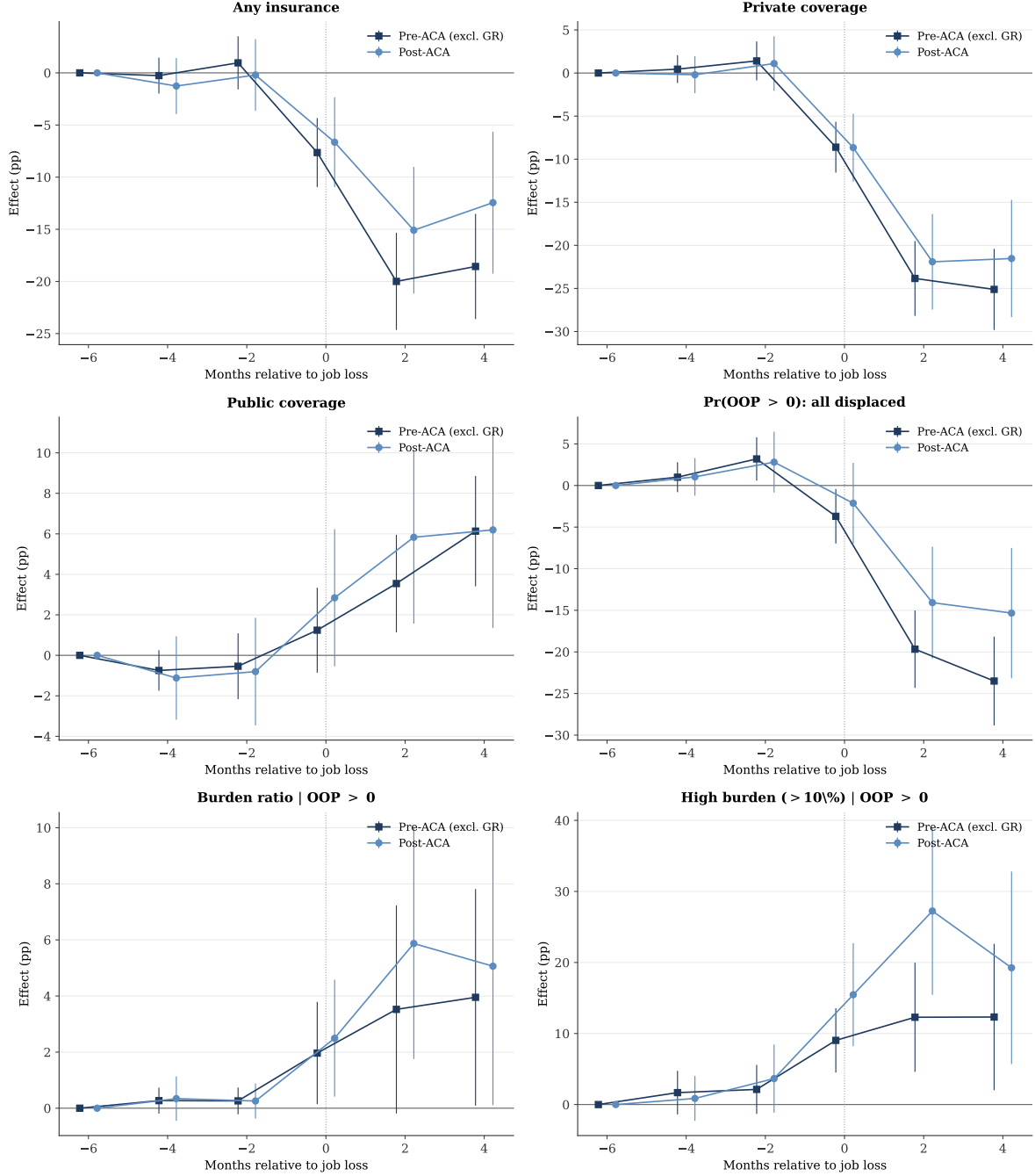


Figure 5: Insurance and burden event study: pre-ACA vs. post-ACA.

Six panels: insurance trio (any insurance, private coverage, public coverage), the Stage 1A extensive premium-paying margin ($\Pr(\text{OOP} > 0)$) on the full panel), and the Stage 2 intensive cost-burden margin (burden ratio and high-burden $> 10\%$ indicator, both conditional on $\text{OOP} > 0$ and household income above the floor). Coefficients from Equation (1) with 95% confidence intervals clustered at the person level. Window restricted to $k \in \{-4, -2, 0, +2, +4\}$ to focus on the causal window. Per- k ACA mitigation values (Post–Pre) and significance stars are reported in Table 3. Stage 2 panels are estimated on the within-payer subsample described in Section 3.3. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$.

Table 3: Main event study: 6 outcomes (insurance trio + two-part burden block), $k \in \{-4, -2, 0, +2, +4\}$, with per- k ACA mitigation

Outcome	Era	N	$k = -4$	$k = -2$	$k = +0$	$k = +2$	$k = +4$	Baseline
Any insurance	Pre-ACA	1,703	-0.003 (0.009)	+0.010 (0.013)	-0.076*** (0.017)	-0.200*** (0.024)	-0.186*** (0.026)	0.659
	Post-ACA	608	-0.013 (0.014)	-0.002 (0.018)	-0.067*** (0.022)	-0.151*** (0.031)	-0.124*** (0.035)	0.763
	Mitigation		-0.010 (0.016)	-0.012 (0.022)	+0.010 (0.028)	+0.049 (0.039)	+0.061 (0.043)	
Private coverage	Pre-ACA	1,703	+0.004 (0.008)	+0.014 (0.012)	-0.086*** (0.015)	-0.238*** (0.022)	-0.251*** (0.024)	0.550
	Post-ACA	608	-0.002 (0.011)	+0.011 (0.016)	-0.087*** (0.020)	-0.219*** (0.028)	-0.215*** (0.035)	0.584
	Mitigation		-0.007 (0.014)	-0.003 (0.020)	-0.001 (0.025)	+0.019 (0.036)	+0.036 (0.042)	
Public coverage	Pre-ACA	1,703	-0.007 (0.005)	-0.005 (0.008)	+0.012 (0.011)	+0.035*** (0.012)	+0.061*** (0.014)	0.117
	Post-ACA	608	-0.011 (0.011)	-0.008 (0.014)	+0.028* (0.017)	+0.058*** (0.022)	+0.062** (0.025)	0.216
	Mitigation		-0.004 (0.012)	-0.003 (0.016)	+0.016 (0.020)	+0.023 (0.025)	+0.001 (0.028)	
Premium-paying margin (Stage 1A)	Pre-ACA	1,703	+0.010 (0.009)	+0.032** (0.013)	-0.037** (0.017)	-0.197*** (0.024)	-0.235*** (0.027)	0.440
	Post-ACA	608	+0.010 (0.012)	+0.028 (0.019)	-0.021 (0.025)	-0.141*** (0.034)	-0.153*** (0.040)	0.433
	Mitigation		+0.000 (0.015)	-0.004 (0.023)	+0.016 (0.030)	+0.056 (0.042)	+0.082* (0.048)	
Burden ratio (Stage 2, intensive)	Pre-ACA	763	+0.003 (0.002)	+0.003 (0.002)	+0.020** (0.009)	+0.035* (0.019)	+0.040** (0.020)	0.072
	Post-ACA	294	+0.003 (0.004)	+0.003 (0.003)	+0.025** (0.011)	+0.059*** (0.021)	+0.051** (0.025)	0.072
	Mitigation		+0.001 (0.005)	-0.000 (0.004)	+0.005 (0.014)	+0.023 (0.028)	+0.011 (0.032)	
High-burden share (>10%, Stage 2)	Pre-ACA	763	+0.017 (0.016)	+0.021 (0.018)	+0.090*** (0.023)	+0.123*** (0.039)	+0.123** (0.053)	0.129
	Post-ACA	294	+0.009 (0.016)	+0.037 (0.025)	+0.155*** (0.037)	+0.272*** (0.060)	+0.192*** (0.069)	0.194
	Mitigation		-0.008 (0.022)	+0.015 (0.030)	+0.064 (0.044)	+0.150** (0.072)	+0.069 (0.087)	

Each outcome panel reports three rows: pre-ACA event-study coefficients across $k \in \{-4, -2, 0, +2, +4\}$, post-ACA coefficients at the same bins, and per- k ACA mitigation $\hat{\Delta}_k = \hat{\beta}_k^{\text{Post}} - \hat{\beta}_k^{\text{Pre}}$ with significance from contrast z -tests using the full variance-covariance matrix (equation (11)). The N column reports unique treated workers in each era's GR-excluded estimation sample (note: this differs from the headcount in Table 2 due to GR-period exclusion and listwise missingness on stage-specific estimating-equation variables); matched controls are paired one-to-one with treated workers within era. Tables 7 and the heterogeneity tables report different N definitions consistent with their per-stratum samples; the N definition is given in each table's footer. Insurance outcomes (panels 1–3) and the Stage 1A extensive premium-paying margin (panel 4, $\text{Pr}(\text{OOP} > 0)$) use the full matched panel; the Stage 2 intensive cost-burden margin (panels 5–6, burden ratio and high-burden indicator) is estimated on the within-payer subsample (private coverage + positive premium + income above floor; see Section 3.3). Coefficients are in proportions; a value of -0.200 means a 20.0pp drop relative to baseline. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

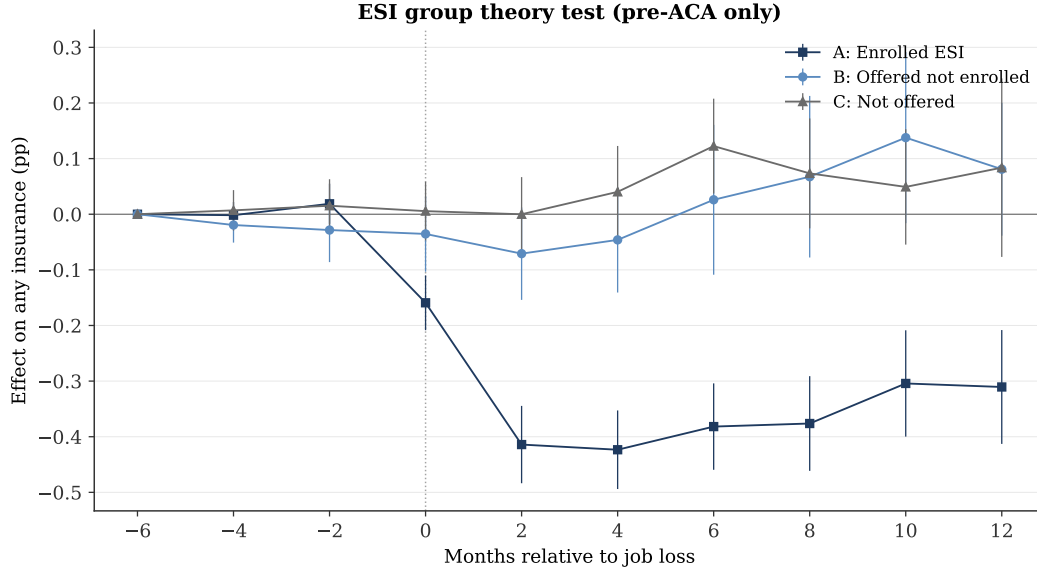


Figure 6: Pre-ACA displacement effect by baseline employer coverage status (theory test)

Pre-ACA-only event study for any-insurance coverage, split into three groups by baseline employer-coverage status: (A) employer-coverage holder at $k = -6$, (B) offered but not taken (likely on a spouse's plan), (C) not offered. The monotone ordering of displacement effects across the three groups is the qualitative prediction of the [Madrian \(1994\)](#) job-lock mechanism.

Table 4: Coverage theory test: any-insurance event study by baseline coverage status, pre-ACA only

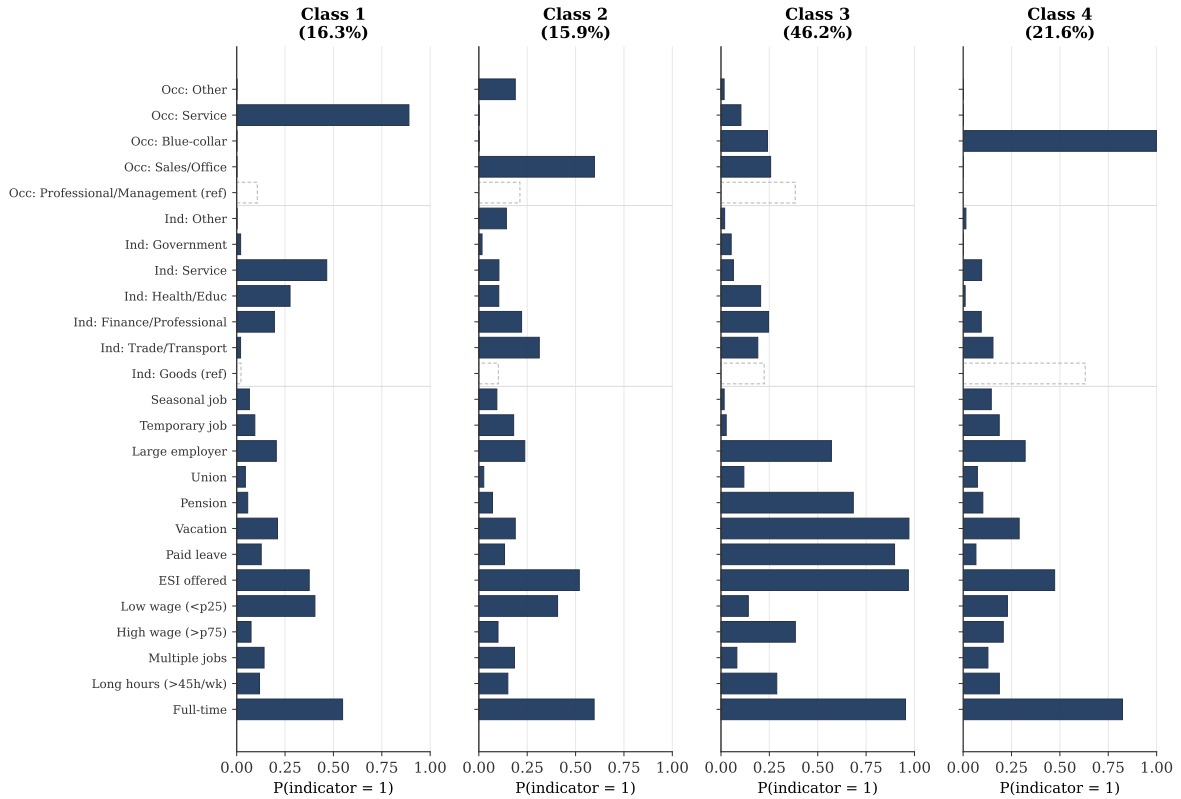
Baseline coverage status	<i>N</i> persons	$k = -4$	$k = -2$	$k = +0$	$k = +2$	$k = +4$
A: Enrolled ESI	760	-0.002 (0.012)	+0.019 (0.018)	-0.159*** (0.025)	-0.414*** (0.035)	-0.423*** (0.036)
B: Offered not enrolled	410	-0.019 (0.016)	-0.028 (0.029)	-0.035 (0.035)	-0.071* (0.042)	-0.046 (0.048)
C: Not offered	533	+0.007 (0.019)	+0.015 (0.024)	+0.006 (0.027)	-0.000 (0.034)	+0.040 (0.042)

Notes. Pre-ACA event-study coefficients on any-insurance coverage, estimated separately within each baseline coverage group at $k = -6$ from equation (1). The three groups are defined by whether the worker was enrolled in own employer-sponsored insurance (Group A), offered employer coverage but not enrolled (Group B, typically covered through a spouse's plan), or not offered employer coverage at all (Group C). The much larger displacement effect in Group A than in Groups B and C is the qualitative prediction of the [Madrian \(1994\)](#) job-lock mechanism: workers with the most to lose lose the most. Standard errors clustered at the person level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5: LCA fit statistics by K

K	LL	BIC	Entropy	Min AvePP	Min Class %
2	−34,731.5	69,832.6	0.869	0.963	46.8%
3	−33,872.9	68,304.1	0.910	0.956	20.8%
4	−33,279.4	67,306.0	0.906	0.917	15.8%
5	−32,787.2	66,510.5	0.925	0.945	3.5%
6	−32,290.5	65,706.1	0.933	0.934	3.4%
7	−31,866.9	65,047.7	0.939	0.937	3.4%

Latent-class fit statistics from the EM algorithm with 25 random starts per K . Information criteria fall monotonically through $K = 7$. Following [Brauner et al. \(2019\)](#), K selection combines information criteria, classification quality (entropy and minimum AvePP), minimum class size, and substantive interpretability rather than minimizing BIC alone. The selected $K = 4$ (bold) is the largest K at which the smallest class remains substantively interpretable on the criteria [Brauner et al. \(2019\)](#) apply and [Masyn \(2013\)](#) discusses qualitatively as a class-collapsing diagnostic; entropy 0.906 and minimum AvePP 0.917 are both well above conventional thresholds.

**Figure 7: Latent-class profiles, $K = 4$.**

Item-response probabilities π_{jk} across the 23 baseline indicators for each of the four classes, with reference-category probabilities (Goods, Professional/Management) computed as $1 - \sum_j \pi_{jk}$ within each dummy set and shown as the top row of each industry / occupation block. Tier labels are generated from the profile by the largest-gap rule on baseline employer-coverage offer; see Appendix D.

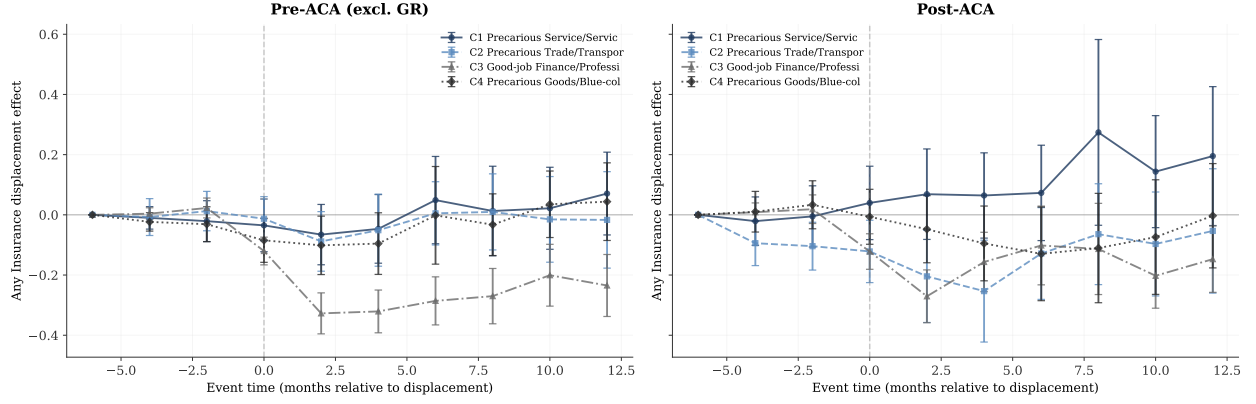


Figure 8: Per-class event study: any-insurance coverage, by era

Event-study trajectories $\hat{\beta}_k^{\text{era}}$ for any-insurance coverage from equation (1), estimated separately within each latent class and era. The divergence between Pre and Post trajectories is the per-class ACA mitigation: visibly positive for C3 (Good-job Professional), weakly positive for C1 (Service) and C4 (Goods and Blue-collar) in point estimate but mostly statistically insignificant; visibly negative throughout the event-time window for C2 (Trade and Sales), interpreted as cohort-level coverage decline rather than ACA mitigation (§5.4.2).

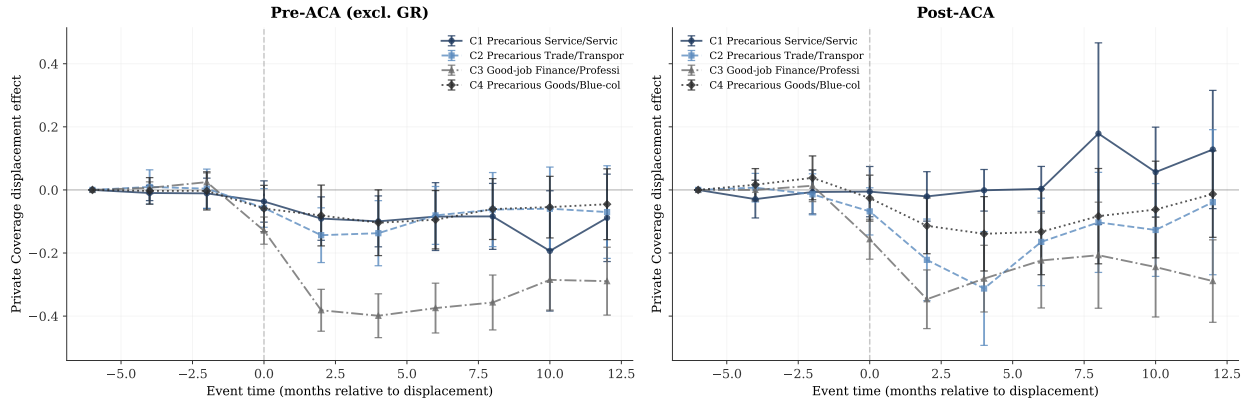


Figure 9: Per-class event study: private coverage, by era

Per-class event-study coefficients for private coverage. Same panel structure as Figure 8.

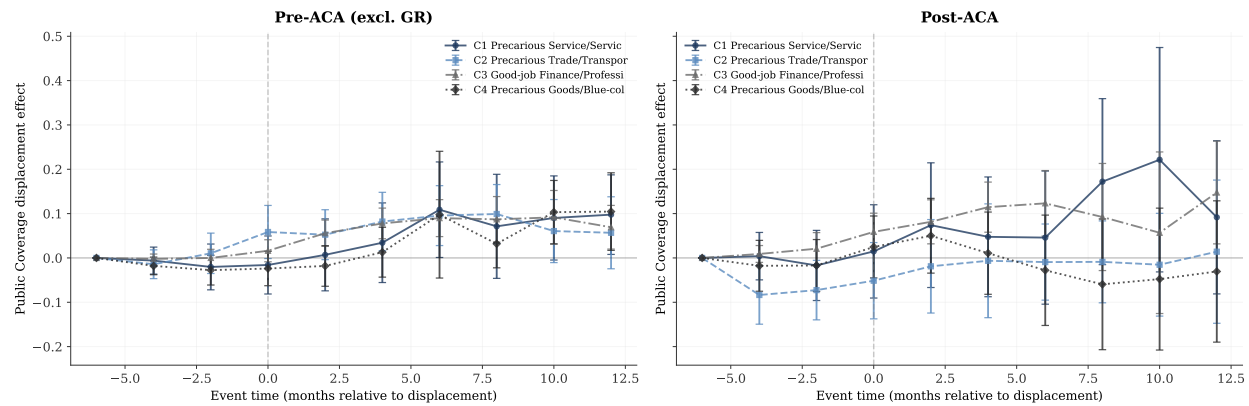


Figure 10: Per-class event study: public coverage, by era

Per-class event-study coefficients for public coverage (Medicaid, Medicare). The super-class joint test on this outcome does not reject homogeneity, consistent with Medicaid eligibility being income-determined rather than class-determined.

Table 6: Per-class event study: insurance trio (any, private, public) at $k \in \{-4, -2, 0, +2, +4\}$, with per- k ACA mitigation

Class	Era	N (T+C)	$k = -4$	$k = -2$	$k = +0$	$k = +2$	$k = +4$
1. Any insurance							
C1 Precarious S	Pre-ACA	578	-0.010 (0.019)	-0.021 (0.035)	-0.035 (0.045)	-0.066 (0.051)	-0.047 (0.058)
	Post-ACA		-0.021 (0.041)	-0.005 (0.052)	+0.040 (0.062)	+0.069 (0.077)	+0.064 (0.072)
	Mitigation		-0.011 (0.045)	+0.016 (0.062)	+0.075 (0.077)	+0.134 (0.092)	+0.111 (0.093)
C2 Precarious T	Pre-ACA	593	-0.007 (0.031)	+0.012 (0.033)	-0.013 (0.037)	-0.088* (0.051)	-0.051 (0.061)
	Post-ACA		-0.094** (0.038)	-0.104** (0.041)	-0.122** (0.053)	-0.204*** (0.078)	-0.253*** (0.086)
	Mitigation		-0.087* (0.049)	-0.117** (0.052)	-0.109* (0.065)	-0.116 (0.093)	-0.202* (0.106)
C3 Good-job Fin	Pre-ACA	1,704	+0.004 (0.010)	+0.023 (0.017)	-0.120*** (0.023)	-0.327*** (0.035)	-0.321*** (0.036)
	Post-ACA		+0.009 (0.016)	+0.019 (0.024)	-0.122*** (0.030)	-0.271*** (0.045)	-0.157*** (0.050)
	Mitigation		+0.005 (0.019)	-0.004 (0.030)	-0.002 (0.038)	+0.057 (0.057)	+0.164*** (0.062)
C4 Precarious G	Pre-ACA	790	-0.023 (0.016)	-0.031 (0.030)	-0.085** (0.037)	-0.101** (0.050)	-0.096* (0.052)
	Post-ACA		+0.010 (0.034)	+0.033 (0.041)	-0.006 (0.047)	-0.048 (0.057)	-0.095 (0.063)
	Mitigation		+0.034 (0.038)	+0.065 (0.050)	+0.078 (0.060)	+0.054 (0.075)	+0.001 (0.082)
2. Private coverage							
C1 Precarious S	Pre-ACA	578	-0.010 (0.012)	-0.011 (0.025)	-0.037 (0.033)	-0.091*** (0.035)	-0.100** (0.041)
	Post-ACA		-0.029 (0.030)	-0.007 (0.035)	-0.005 (0.041)	-0.021 (0.040)	-0.001 (0.034)
	Mitigation		-0.019 (0.033)	+0.004 (0.043)	+0.031 (0.053)	+0.070 (0.053)	+0.098* (0.053)
C2 Precarious T	Pre-ACA	593	+0.010 (0.028)	+0.004 (0.032)	-0.057* (0.031)	-0.144*** (0.044)	-0.137*** (0.053)
	Post-ACA		+0.007 (0.023)	-0.014 (0.033)	-0.068* (0.038)	-0.222*** (0.066)	-0.313*** (0.092)
	Mitigation		-0.002 (0.036)	-0.018 (0.046)	-0.011 (0.049)	-0.078 (0.080)	-0.175* (0.106)
C3 Good-job Fin	Pre-ACA	1,704	+0.007 (0.010)	+0.025* (0.015)	-0.129*** (0.022)	-0.382*** (0.034)	-0.399*** (0.035)
	Post-ACA		+0.000 (0.016)	+0.013 (0.026)	-0.156*** (0.032)	-0.347*** (0.047)	-0.281*** (0.054)
	Mitigation		-0.007 (0.018)	-0.011 (0.030)	-0.027 (0.039)	+0.035 (0.058)	+0.118* (0.065)
C4 Precarious G	Pre-ACA	790	-0.003 (0.021)	-0.003 (0.031)	-0.059 (0.038)	-0.081* (0.049)	-0.104* (0.053)
	Post-ACA		+0.016 (0.026)	+0.038 (0.035)	-0.026 (0.037)	-0.114** (0.045)	-0.139** (0.060)
	Mitigation		+0.019 (0.034)	+0.041 (0.047)	+0.033 (0.053)	-0.033 (0.067)	-0.035 (0.080)
3. Public coverage							
C1 Precarious S	Pre-ACA	578	-0.006 (0.016)	-0.020 (0.026)	-0.016 (0.033)	+0.007 (0.041)	+0.034 (0.046)
	Post-ACA		+0.004 (0.027)	-0.017 (0.040)	+0.015 (0.054)	+0.074 (0.072)	+0.048 (0.069)
	Mitigation		+0.010 (0.031)	+0.003 (0.048)	+0.031 (0.063)	+0.067 (0.083)	+0.013 (0.083)
C2 Precarious T	Pre-ACA	593	-0.014 (0.017)	+0.010 (0.023)	+0.058* (0.031)	+0.052* (0.029)	+0.082** (0.034)
	Post-ACA		-0.084** (0.034)	-0.073** (0.034)	-0.051 (0.044)	-0.019 (0.054)	-0.006 (0.065)
	Mitigation		-0.069* (0.037)	-0.083** (0.041)	-0.110** (0.053)	-0.071 (0.061)	-0.088 (0.074)
C3 Good-job Fin	Pre-ACA	1,704	-0.002 (0.006)	-0.000 (0.010)	+0.016 (0.013)	+0.056*** (0.015)	+0.078*** (0.018)
	Post-ACA		+0.009 (0.010)	+0.021 (0.019)	+0.059*** (0.022)	+0.081*** (0.025)	+0.115*** (0.029)
	Mitigation		+0.011 (0.011)	+0.021 (0.021)	+0.043* (0.025)	+0.026 (0.030)	+0.037 (0.034)
C4 Precarious G	Pre-ACA	790	-0.018* (0.010)	-0.028 (0.017)	-0.024 (0.020)	-0.018 (0.023)	+0.013 (0.029)
	Post-ACA		-0.018 (0.029)	-0.017 (0.030)	+0.025 (0.036)	+0.050 (0.043)	+0.011 (0.047)
	Mitigation		+0.000 (0.031)	+0.010 (0.034)	+0.049 (0.041)	+0.068 (0.049)	-0.002 (0.055)

Notes. Per-class event study on the insurance trio (any insurance, private coverage, public coverage) at $k \in \{-4, -2, 0, +2, +4\}$. Three panels (one per outcome), four rows per panel (one per LCA class), three lines per (class, outcome) cell: pre-ACA / post-ACA / per- k ACA mitigation with significance stars from contrast z -tests using the full variance-covariance matrix. *The Trade and Sales (C2) any-insurance and public-coverage rows exhibit a downward post-ACA cohort trajectory at every event-time horizon including pre-displacement (post-ACA-only pre-trend joint Wald $p < 0.05$ on both); the negative point estimates at post-displacement horizons in this class are not interpreted as causal ACA effects (§5.4.2).* The per-class affordability margin is reported at the super-class level in Table 7. Standard errors clustered at the person level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

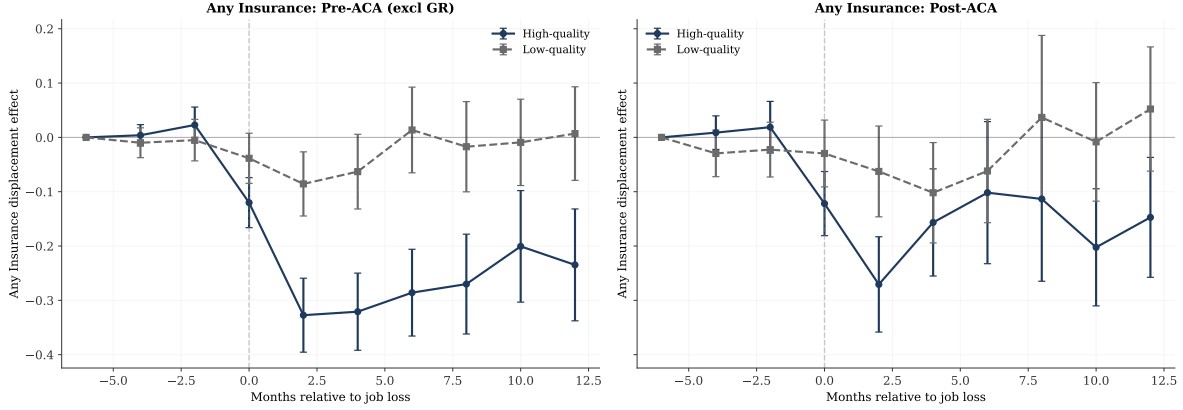


Figure 11: Super-class event study: any insurance (Good-job vs. Precarious)

Event-study trajectories for the two super-classes (Good-job = C3; Precarious = C1+C2+C4) on the any-insurance outcome. The per-horizon HQ-vs-LQ contrast at $k = +4$ is +15.5pp, individually significant at 5%.

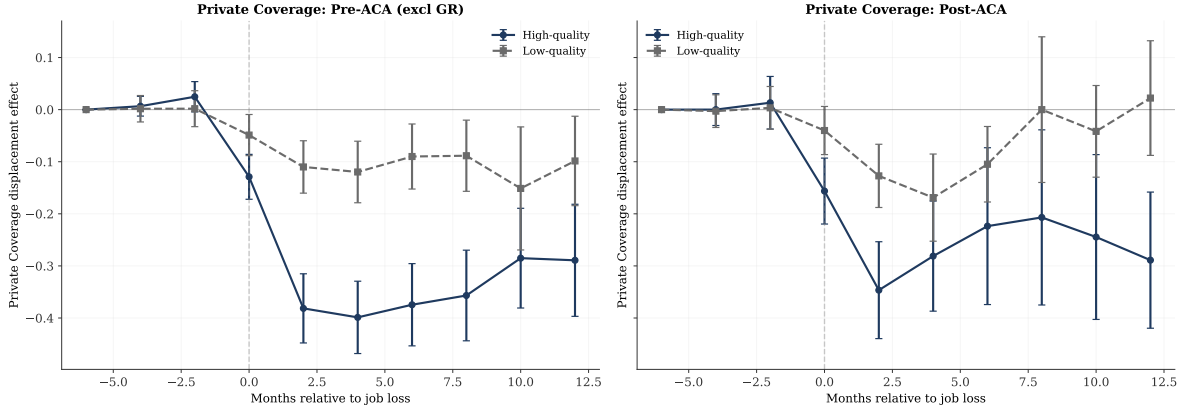


Figure 12: Super-class event study: private coverage (Good-job vs. Precarious)

Private-coverage event study by super-class and era. The HQ-vs-LQ joint test rejects parallel ACA mitigation on this outcome ($\chi^2_9 = 18.13$, $p = 0.034$; Table 7, Panel D, row 2), the second of the two extensive-margin DDD rejections.

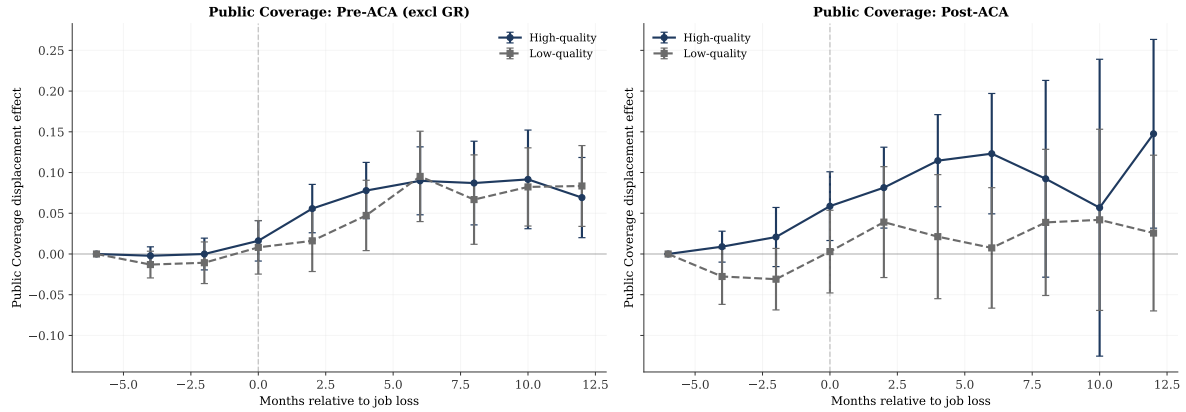


Figure 13: Super-class event study: public coverage (Good-job vs. Precarious)

Public-coverage event study by super-class and era. The HQ-vs-LQ joint test does not reject parallel mitigation on this outcome ($\chi^2_9 = 10.49$, $p = 0.313$), consistent with Medicaid eligibility being income-determined rather than worker-type-determined.

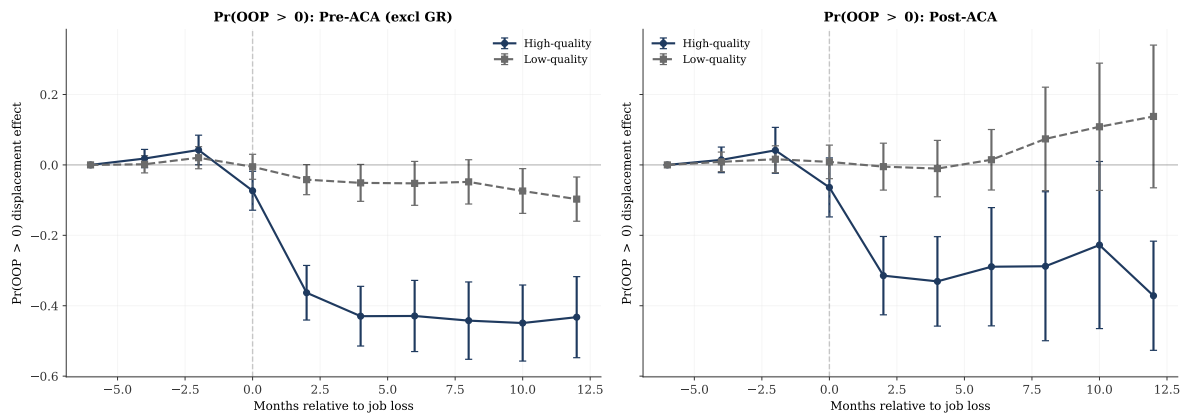


Figure 14: Super-class event study: premium-paying margin (extensive)

Event-study trajectories on the extensive premium-paying margin, whether the worker pays any out-of-pocket premium ($\Pr(\text{OOP} > 0)$), estimated on the full matched panel within each super-class. This is the Stage 1A outcome of the two-part hurdle decomposition described in Section 4.5. The HQ-vs-LQ joint test rejects parallel ACA mitigation on this outcome ($\chi^2_9 = 18.39$, $p = 0.031$; Table 7, Panel D, row 4), indicating the ACA's premium-paying coverage gains were concentrated in the high-quality super-class.

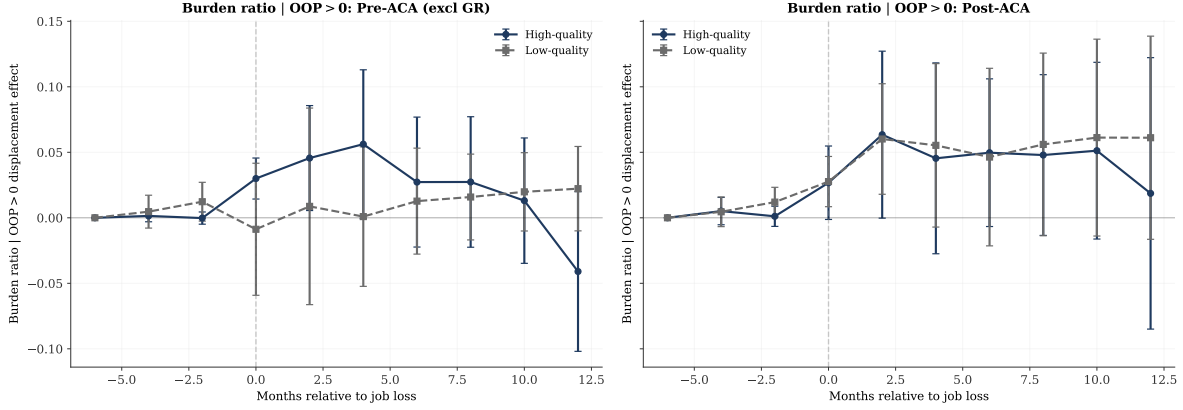


Figure 15: Super-class event study: burden ratio (intensive, among premium-payers)

Event-study trajectories on the intensive cost-burden margin: the conditional premium-burden ratio among workers with private coverage, a positive out-of-pocket premium, and household income above the floor (Stage 2 of the two-part hurdle decomposition). The within-payer ACA mitigation point estimate is +1.8pp at $k = +2$ within HQ and +5.1pp at $k = +2$ within LQ; the HQ-vs-LQ joint test does not reject parallel mitigation ($\chi^2_9 = 8.96$, $p = 0.441$). Selection caveat: the within-payer sample is more selected for the LQ super-class than for HQ.

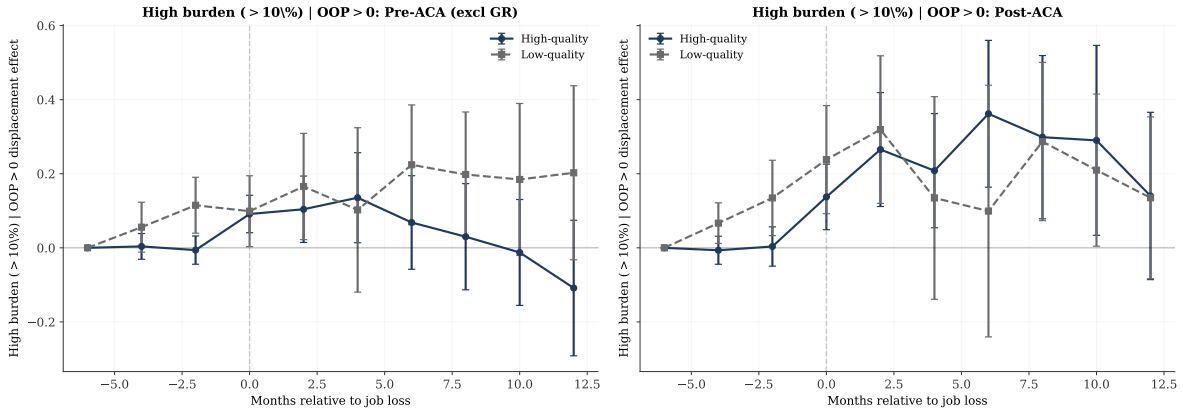


Figure 16: Super-class event study: high-burden share (>10%) (intensive, among premium-payers)

Event-study trajectories on the high-burden indicator within payers, by super-class and era. The within-payer ACA mitigation point estimate is +16.1pp at $k = +2$ within HQ (significant at 10%) and +15.4pp at $k = +2$ within LQ (similar magnitude, less precisely identified, $p > 0.10$ individually). The HQ-vs-LQ joint test does not reject parallel mitigation ($\chi^2_9 = 8.91$, $p = 0.446$); the within-payer mitigation pattern is broadly parallel across super-classes.

Table 7: Super-class event study and joint heterogeneity test: 6 outcomes (insurance trio + two-part burden block)

Era	<i>N</i> persons	<i>k</i> = − 4	<i>k</i> = − 2	<i>k</i> = + 0	<i>k</i> = + 2	<i>k</i> = + 4
Panel A. Good-job super-class (HQ)						
1. Any insurance						
Pre-ACA	1,704	+0.004 (0.010)	+0.023 (0.017)	−0.120*** (0.023)	−0.327*** (0.035)	−0.321*** (0.036)
Post-ACA		+0.009 (0.016)	+0.019 (0.024)	−0.122*** (0.030)	−0.271*** (0.045)	−0.157*** (0.050)
Mitigation		+0.005 (0.019)	−0.004 (0.030)	−0.002 (0.038)	+0.057 (0.057)	+0.164*** (0.062)
2. Private coverage						
Pre-ACA	1,704	+0.007 (0.010)	+0.025* (0.015)	−0.129*** (0.022)	−0.382*** (0.034)	−0.399*** (0.035)
Post-ACA		+0.000 (0.016)	+0.013 (0.026)	−0.156*** (0.032)	−0.347*** (0.047)	−0.281*** (0.054)
Mitigation		−0.007 (0.018)	−0.011 (0.030)	−0.027 (0.039)	+0.035 (0.058)	+0.118* (0.065)
3. Public coverage						
Pre-ACA	1,704	−0.002 (0.006)	−0.000 (0.010)	+0.016 (0.013)	+0.056*** (0.015)	+0.078*** (0.018)
Post-ACA		+0.009 (0.010)	+0.021 (0.019)	+0.059*** (0.022)	+0.081*** (0.025)	+0.115*** (0.029)
Mitigation		+0.011 (0.011)	+0.021 (0.021)	+0.043* (0.025)	+0.026 (0.030)	+0.037 (0.034)
4. Premium-paying margin (Stage 1A)						
Pre-ACA	1,704	+0.018 (0.013)	+0.042** (0.021)	−0.073*** (0.028)	−0.363*** (0.040)	−0.430*** (0.043)
Post-ACA		+0.014 (0.019)	+0.041 (0.033)	−0.064 (0.043)	−0.315*** (0.057)	−0.331*** (0.065)
Mitigation		−0.004 (0.023)	−0.001 (0.040)	+0.009 (0.051)	+0.049 (0.069)	+0.099 (0.078)
5. Burden ratio (Stage 2, intensive)						
Pre-ACA	1,192	+0.001 (0.002)	−0.000 (0.002)	+0.030*** (0.008)	+0.046** (0.020)	+0.056* (0.029)
Post-ACA		+0.005 (0.005)	+0.001 (0.004)	+0.027* (0.014)	+0.063* (0.033)	+0.045 (0.037)
Mitigation		+0.004 (0.006)	+0.001 (0.005)	−0.003 (0.016)	+0.018 (0.038)	−0.011 (0.047)
6. High-burden share (>10%, Stage 2)						
Pre-ACA	1,192	+0.004 (0.018)	−0.006 (0.019)	+0.091*** (0.026)	+0.104** (0.046)	+0.135** (0.062)
Post-ACA		−0.007 (0.019)	+0.004 (0.027)	+0.137*** (0.045)	+0.265*** (0.078)	+0.208*** (0.079)
Mitigation		−0.010 (0.026)	+0.010 (0.033)	+0.046 (0.052)	+0.161* (0.091)	+0.073 (0.100)
Panel B. Precarious super-class (LQ)						
1. Any insurance						
Pre-ACA	1,961	−0.010 (0.014)	−0.005 (0.019)	−0.038 (0.024)	−0.086*** (0.030)	−0.063* (0.035)
Post-ACA		−0.029 (0.022)	−0.023 (0.026)	−0.030 (0.031)	−0.063 (0.043)	−0.102** (0.047)
Mitigation		−0.020 (0.026)	−0.017 (0.032)	+0.009 (0.039)	+0.023 (0.052)	−0.039 (0.059)
2. Private coverage						
Pre-ACA	1,961	+0.002 (0.013)	+0.002 (0.018)	−0.049** (0.020)	−0.110*** (0.026)	−0.120*** (0.030)
Post-ACA		−0.003 (0.016)	+0.003 (0.021)	−0.040* (0.024)	−0.127*** (0.031)	−0.169*** (0.043)
Mitigation		−0.005 (0.021)	+0.002 (0.027)	+0.009 (0.031)	−0.017 (0.040)	−0.049 (0.052)
3. Public coverage						
Pre-ACA	1,961	−0.013 (0.008)	−0.011 (0.013)	+0.008 (0.017)	+0.016 (0.019)	+0.047** (0.022)
Post-ACA		−0.028 (0.017)	−0.031 (0.019)	+0.003 (0.026)	+0.039 (0.035)	+0.021 (0.039)
Mitigation		−0.015 (0.019)	−0.020 (0.023)	−0.005 (0.031)	+0.023 (0.040)	−0.026 (0.045)
4. Premium-paying margin (Stage 1A)						
Pre-ACA	1,961	+0.002 (0.013)	+0.020 (0.016)	−0.005 (0.018)	−0.042* (0.022)	−0.051* (0.027)
Post-ACA		+0.009 (0.014)	+0.016 (0.019)	+0.009 (0.024)	−0.005 (0.034)	−0.010 (0.041)
Mitigation		+0.007 (0.019)	−0.004 (0.025)	+0.014 (0.030)	+0.037 (0.040)	+0.041 (0.049)
5. Burden ratio (Stage 2, intensive)						
Pre-ACA	471	+0.005 (0.006)	+0.012 (0.008)	−0.009 (0.026)	+0.009 (0.038)	+0.001 (0.027)
Post-ACA		+0.005 (0.006)	+0.012** (0.006)	+0.028*** (0.010)	+0.060*** (0.022)	+0.055* (0.032)
Mitigation		−0.000 (0.009)	−0.000 (0.010)	+0.036 (0.028)	+0.051 (0.044)	+0.054 (0.042)
6. High-burden share (>10%, Stage 2)						
Pre-ACA	471	+0.056 (0.034)	+0.115*** (0.039)	+0.099** (0.049)	+0.166** (0.073)	+0.102 (0.113)

(continued on next page)

(Table 7, continued)

Era	N persons	$k = -4$	$k = -2$	$k = +0$	$k = +2$	$k = +4$
Post-ACA		+0.067** (0.028)	+0.135*** (0.052)	+0.238*** (0.074)	+0.319*** (0.102)	+0.135 (0.140)
Mitigation		+0.011 (0.044)	+0.020 (0.065)	+0.139 (0.089)	+0.154 (0.125)	+0.032 (0.180)
Panel C. HQ-vs-LQ mitigation contrasts (DDD)						
$Per\text{-}k$ DDD contrast: $\hat{\phi}_k^{Post} - \hat{\phi}_k^{Pre}$ at each k . Joint test in Panel D.						
1. Any insurance		+0.013 (0.025)	−0.002 (0.030)	−0.021 (0.041)	+0.087 (0.063)	+0.155** (0.067)
2. Private coverage		−0.018 (0.022)	−0.040 (0.028)	−0.054 (0.037)	+0.054 (0.060)	+0.095 (0.066)
3. Public coverage		+0.028* (0.017)	+0.047** (0.023)	+0.040 (0.031)	+0.033 (0.039)	+0.062 (0.046)
4. Premium-paying margin (Stage 1A)		−0.005 (0.024)	−0.021 (0.036)	−0.065 (0.045)	−0.013 (0.066)	−0.004 (0.070)
5. Burden ratio (Stage 2)		+0.010 (0.010)	+0.009 (0.009)	−0.023 (0.030)	−0.004 (0.057)	−0.041 (0.061)
6. High-burden share (Stage 2)		+0.026 (0.033)	+0.047 (0.044)	+0.012 (0.073)	+0.094 (0.135)	+0.030 (0.151)
Panel D. Joint Wald test: HQ-vs-LQ mitigation, all 9 event-time bins						
H_0 : $\phi_k^{Post} = \phi_k^{Pre}$ jointly across $k \in \{-4, -2, 0, +2, +4, +6, +8, +10, +12\}$.						
Outcome		χ^2_9	d.f.	p -value	Reject H_0 ?	
1. Any insurance		12.63	9	0.1799	No	
2. Private coverage		18.13**	9	0.0337	Yes (5%)	
3. Public coverage		10.49	9	0.3126	No	
4. Premium-paying margin (Stage 1A)		18.39**	9	0.0309	Yes (5%)	
5. Burden ratio (Stage 2)		8.96	9	0.4406	No	
6. High-burden share (Stage 2)		8.91	9	0.4459	No	

Notes. Super-class event studies, HQ-vs-LQ mitigation contrasts, and joint Wald tests for heterogeneous ACA mitigation on six outcomes: insurance trio (any insurance, private, public), the Stage 1A extensive premium-paying margin ($\Pr(\text{OOP} > 0)$ on the full panel), and the Stage 2 intensive cost-burden margin (burden ratio and high-burden indicator, both estimated conditional on $\text{OOP} > 0$ and household income above the floor). **Panels A and B** report the per-era event-study coefficients and within-super-class mitigation ($\hat{\beta}_k^{Post} - \hat{\beta}_k^{Pre}$). **Panel C** reports the per- k HQ-vs-LQ mitigation contrasts from the specification in equation (5). **Panel D** reports the joint χ^2_9 Wald test on the 9 mitigation contrasts across all event-time bins; this answers the formal policy question “is the ACA mitigation different in HQ vs. LQ super-classes?” Insurance outcomes (rows 1–3) and Stage 1A (row 4) use the full super-class panel; Stage 2 outcomes (rows 5–6) are estimated on the within-payer subsample (private coverage + positive premium + income above floor) within super-class, with the within-payer selection more severe for the LQ super-class than for HQ. Standard errors clustered at the person level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

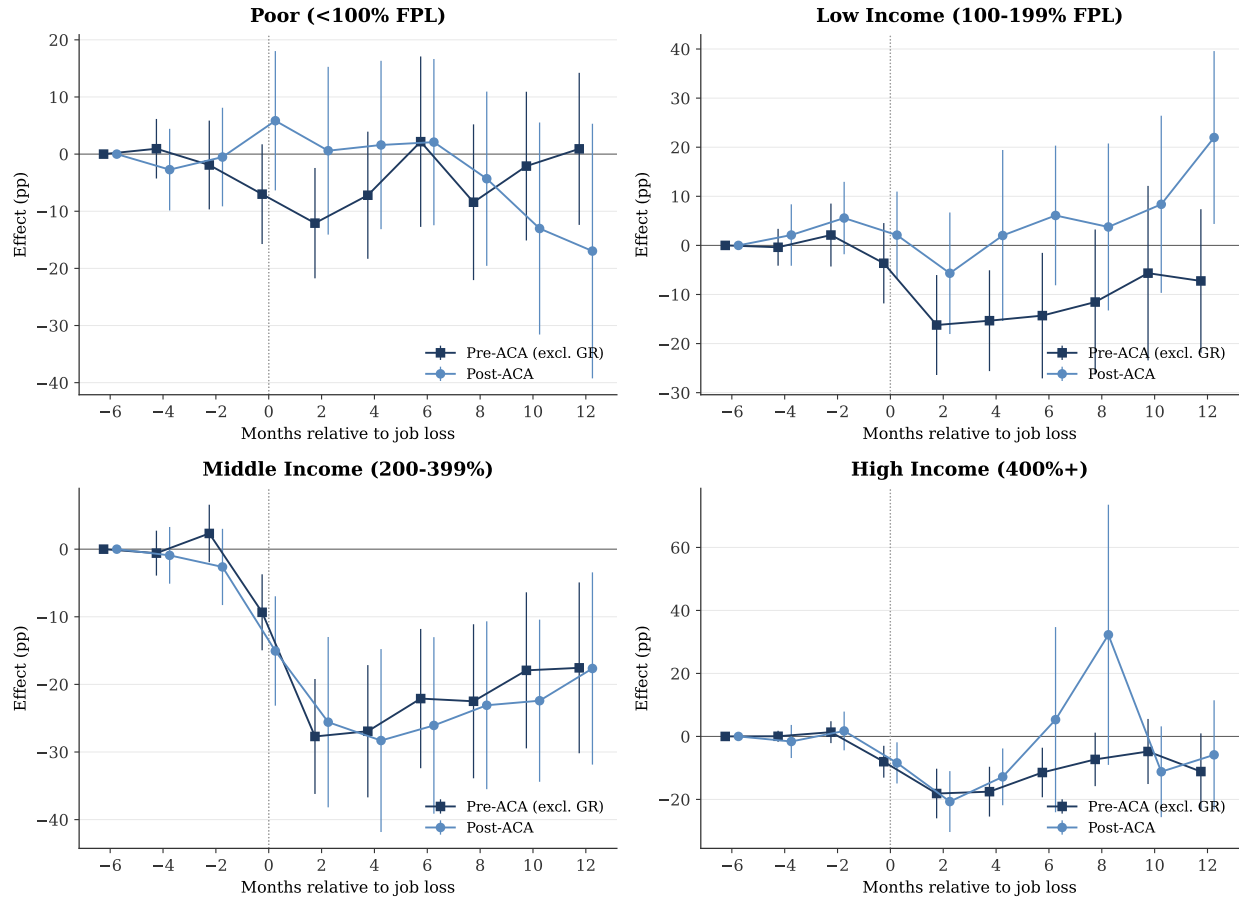


Figure 17: Insurance event study by baseline income group (any insurance)

Heterogeneity in the any-insurance displacement effect by baseline income group. Each subplot estimates equation (1) on a separate income bin and overlays Pre-ACA and Post-ACA trajectories. The largest pre/post divergence falls in the Low Income (100–199% FPL) bin, consistent with Marketplace subsidies and Medicaid expansion reaching this population.

Table 8: Income heterogeneity: any-insurance event study, $k \in \{-4, -2, 0, +2, +4\}$, with per- k ACA mitigation

Income group	Era	N persons	$k = -4$	$k = -2$	$k = +0$	$k = +2$	$k = +4$
Poor (<100% FPL)	Pre-ACA	346	+0.009 (0.027)	-0.019 (0.040)	-0.070 (0.045)	-0.121** (0.049)	-0.072 (0.057)
	Post-ACA	145	-0.027 (0.036)	-0.005 (0.044)	+0.058 (0.062)	+0.006 (0.075)	+0.016 (0.075)
	Mitigation		-0.037 (0.045)	+0.014 (0.059)	+0.129* (0.076)	+0.127 (0.090)	+0.088 (0.094)
Low Income (100-199% FPL)	Pre-ACA	583	-0.004 (0.019)	+0.021 (0.033)	-0.036 (0.042)	-0.162*** (0.052)	-0.153*** (0.052)
	Post-ACA	172	+0.021 (0.032)	+0.056 (0.038)	+0.021 (0.045)	-0.057 (0.063)	+0.020 (0.089)
	Mitigation		+0.025 (0.037)	+0.035 (0.050)	+0.057 (0.061)	+0.105 (0.082)	+0.174* (0.103)
Middle Income (200-399%)	Pre-ACA	688	-0.006 (0.017)	+0.023 (0.022)	-0.093*** (0.029)	-0.277*** (0.043)	-0.269*** (0.050)
	Post-ACA	233	-0.009 (0.021)	-0.026 (0.029)	-0.151*** (0.041)	-0.256*** (0.064)	-0.283*** (0.069)
	Mitigation		-0.003 (0.027)	-0.050 (0.036)	-0.057 (0.050)	+0.021 (0.078)	-0.014 (0.085)
High Income (400%+)	Pre-ACA	425	+0.000 (0.009)	+0.013 (0.018)	-0.080*** (0.026)	-0.181*** (0.040)	-0.175*** (0.040)
	Post-ACA	173	-0.016 (0.027)	+0.017 (0.031)	-0.084** (0.033)	-0.207*** (0.049)	-0.128*** (0.046)
	Mitigation		-0.017 (0.028)	+0.004 (0.036)	-0.004 (0.042)	-0.026 (0.064)	+0.047 (0.061)

Heterogeneity in the any-insurance ACA mitigation effect by baseline household income group. Equation (1) is estimated separately on each of four merged poverty-category income bins (Poor; Low Income 100–199%; Middle; High); the Low Income bin merges the original 100–124% (Near Poor) and 125–199% subbands of the matching cells to dissolve a small-sample pre-trend asymmetry in the post-2014 cohort. Three rows per income group: pre-ACA / post-ACA / per- k ACA mitigation across $k \in \{-4, -2, 0, +2, +4\}$. N is total unique persons (treated + control combined) in each group’s regression. The Low Income (100–199% FPL) bin shows positive mitigation at every post-displacement horizon, reaching +17.4pp at $k = +4$ (significant at 10%). Standard errors clustered at the person level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9: Robustness checks: any insurance at $k = +2$

Specification	Pre-ACA	Post-ACA	Mitigation
(1) Main (3-period, reweighted)	−0.200 (0.024)	−0.151 (0.031)	+0.049
(2) No reweighting	−0.199 (0.022)	−0.150 (0.031)	+0.049
(3) Drop Great Recession	−0.199 (0.024)	−0.151 (0.031)	+0.048
(4) GR in pre-ACA	−0.234 (0.018)	−0.151 (0.031)	+0.083
(5) Drop 2018	−0.198 (0.024)	−0.151 (0.032)	+0.047

Five-specification robustness comparison for the headline any-insurance effect at $k = +2$. The Mitigation column is Post−Pre. Coefficients in proportions; standard errors clustered at the person level in parentheses. Specification (4) classifies the Great Recession years (2008–2010) as part of the pre-ACA period rather than a separate era; the larger pre-ACA sample tightens the pre-ACA estimate, growing the Mitigation magnitude.

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A Data construction details

MEPS data sources

This subsection gives the technical detail behind the brief description of the MEPS-HC files in Section 3.1. The three public-use files I pool are identified by AHRQ “HC- n ” release numbers; the full 2004–2018 mapping is in Table 10. The files are linked at the person level by a harmonized respondent identifier and at the job level, when a private plan is obtained through an employer, by a job index.

The **Full-Year Consolidated file** (FYC) has one record per respondent per calendar year. It carries all demographic and socioeconomic information (age, sex, race, ethnicity, education, marital status, family structure, household income, the poverty ratio, and self-reported health status). For each of the twelve months of the reference year it records private and public coverage status; the three insurance outcomes are constructed from these monthly indicators.

The **Jobs file** is at the person-round-job level. For every job a respondent held during the reference period it records start and end dates, employer size and sector, industry, occupation, weekly hours, the wage, and a set of non-wage benefit indicators including round-specific employer-sponsored insurance offer, paid sick leave, paid vacation, pension enrollment, union membership, and a temporary-job indicator. The reason-for-leaving codes distinguish business closure, layoff, being fired, voluntary quit, retirement, and several other outcomes; the *involuntary-displacement* treatment in this paper is defined as the union of the business-closure, layoff, and being-fired codes.

The **Person Round Plan file** is at the person-round-policyholder-establishment-plan level and contains records only when the respondent is covered by a private plan in that round. It reports the policyholder, the source of coverage, and the out-of-pocket premium. The premium amount is AHRQ-edited and harmonized across the units in which respondents originally reported (weekly, biweekly, monthly, quarterly, annually); a small share of records with extreme reports are edited, and missing values are filled by hot-deck imputation. The premium is zero by construction for policyholders whose entire premium is paid by the employer or another party.

Table 10: MEPS-HC public-use file release numbers, 2004–2018

Year	Full-Year Consolidated (FYC)	Jobs file	Person Round Plan (PRPL)
2004	HC-089	HC-083	HC-088
2005	HC-097	HC-091	HC-095
2006	HC-105	HC-100	HC-103
2007	HC-113	HC-108	HC-111
2008	HC-121	HC-116	HC-119
2009	HC-129	HC-124	HC-127
2010	HC-138	HC-133	HC-136
2011	HC-147	HC-142	HC-145
2012	HC-155	HC-150	HC-153
2013	HC-163	HC-158	HC-161
2014	HC-171	HC-166	HC-169
2015	HC-181	HC-176	HC-179
2016	HC-192	HC-185	HC-191
2017	HC-201	HC-195	HC-200
2018	HC-209	HC-203	HC-208

Notes. AHRQ Medical Expenditure Panel Survey Household Component (MEPS-HC) public-use file release numbers for the three files pooled in the analysis. Files are available from the AHRQ MEPS-HC website (https://meps.ahrq.gov/mepsweb/data_stats/download_data_files.jsp) and are linked at the person level by a harmonized respondent identifier.

Monthly income components for the burden ratio

The monthly household income denominator is built from three person-level monthly income components, summed across all members of the health-insurance eligibility unit. *Labor earnings* in month t are observed when the worker holds an active job and are zero in months in which the worker is not employed. *Annual transfer income* (unemployment compensation, public assistance, Supplemental Security Income, and other regular cash contributions) is allocated to the months of the year in which the recipient was not working. *Annual unearned income* (interest, dividends, private pension and IRA withdrawals, Social Security, veterans’ payments, trust and royalty income, business and farm income, alimony and child support received, and other miscellaneous unearned items) is allocated uniformly across all twelve months of the year. All dollar series are deflated to constant 2016 dollars using the BLS Consumer Price Index for All Urban Consumers ([U.S. Bureau of Labor Statistics, 2024](#)) before the burden ratio is formed. Because the same year-specific deflator applies to numerator and denominator, the burden ratio itself is invariant to inflation adjustment; the deflator only affects level statistics like baseline household income and the dollar floor below.

Floor-binding rate

The dollar floor of \$1,000 per year (\$83 per month) in 2016 dollars binds for under 17% of matched household-months across the analytical sample, well within the range [Selden \(2009\)](#) documents on a comparable working-age sample. The floor binds disproportionately at the post-displacement horizons where labor earnings are lowest. Floor-binding household-months are excluded from the burden-valid sample.

Alternative-floor sensitivity

Replacing the \$1,000 (2016 dollars) floor with the \$1,270 (2016 dollars) floor that corresponds exactly to Selden’s original \$1,000 (2004 dollars) shrinks the burden-valid sample by under 2%. Re-estimating the headline burden specifications on this slightly tighter sample yields per-horizon mitigation estimates that are within 0.005 of the values reported in [Table 3](#) for the burden ratio and within 0.01 for the high-burden indicator; all qualitative significance patterns are preserved.

B Empirical-strategy details

Matching procedure

This subsection gives the technical detail behind the matching procedure summarized in [Section 4.2](#).

Step 1: exact cells. Each treated and candidate-control person-month is assigned to a cell defined by the Cartesian product of four variables measured at the baseline month $k = -6$: calendar year-month; sex; Census region; and income group, defined by baseline poverty category in five bins (Poor below 100% of FPL; Near Poor 100 to 124%; Low Income 125 to 199%; Middle Income 200 to 399%; High Income 400% of FPL or above). Candidate controls are workers with at least six consecutive months of employment ending at the baseline month who do not experience an involuntary separation within the estimation window. The income-heterogeneity stratification reported in [Section 5.3](#) merges the smaller Near Poor and Low Income cells into a single Low Income bin (100–199% FPL); the matching cells themselves retain the original five categories.

Step 2: propensity score within cell. Within each cell I estimate a logit model of the probability of involuntary displacement,

$$\Pr(T_i = 1 \mid \mathbf{W}_i) = \frac{\exp(\mathbf{W}_i' \boldsymbol{\eta})}{1 + \exp(\mathbf{W}_i' \boldsymbol{\eta})}, \quad (6)$$

where $T_i = 1$ indicates a displaced worker and $\mathbf{W}_i$ is a vector of baseline covariates measured at $k = -6$. The covariate specification follows [Schaller and Zerpa \(2026\)](#) in spirit but uses a slightly leaner functional form: $\mathbf{W}_i$ includes demographics (age; race and ethnicity, with female interactions; education category dummies); family structure (married, has children, each interacted with sex); job characteristics (full-time indicator and missing-flag, real monthly earnings, with missing and top-code flags); family income (monthly family income as percent of FPL, with missing flag); and baseline insurance status (private, public). I omit the squared terms in age, earnings, and FPL that [Schaller and Zerpa \(2026\)](#) include; on this matched, exact-cell-stratified sample they provide negligible additional balance and the linear specification is more robust to the few-observation cells. The logit is estimated separately by sex.

Greedy nearest-neighbor matching. I form treated-control pairs by greedy nearest-neighbor matching on $\hat{p}_i$, allowing replacement at the candidate-pool stage but not duplication in the final matched sample. Each treated worker’s ten nearest candidates are identified in a first pass; in a second pass, each treated worker is assigned its closest available candidate, with ties broken randomly and unmatched treated workers reassigned to their next-closest available candidate iteratively. No inter-cell substitution is allowed, preserving the exact-match structure of Step 1.

Cross-era reweighting

This subsection gives the technical detail behind the reweighting procedure summarized in Section 4.3.

I estimate a logit on the matched baseline observations only (one observation per worker at $k = -6$),

$$\Pr(\text{Post}_i = 1 \mid \mathbf{Z}_i) = \frac{\exp(\mathbf{Z}_i' \boldsymbol{\gamma})}{1 + \exp(\mathbf{Z}_i' \boldsymbol{\gamma})}, \quad (7)$$

where $\text{Post}_i = 1$ if the worker’s baseline month is January 2014 or later, and $\mathbf{Z}_i$ collects pre-determined characteristics only: sex, race and ethnicity (with sex interactions), age and age², education, marital status (with sex interaction), family size and family size² (with sex interaction on family size), full-time status (with missing-flag), region (Midwest/South/West dummies), and the same five income bins used for exact matching. Critically, $\mathbf{Z}_i$ excludes baseline earnings and baseline insurance coverage, because those are outcomes of the policy environment.

Let $\hat{p}_i = \hat{\Pr}(\text{Post}_i = 1 \mid \mathbf{Z}_i)$. I assign worker-level inverse-odds weights

$$w_i = \begin{cases} 1, & \text{if } B_i \geq \text{January 2014,} \\ \hat{p}_i / (1 - \hat{p}_i), & \text{if } B_i < \text{January 2014,} \end{cases} \quad (8)$$

where B_i is worker i 's baseline month. Each matched pair receives a single pair weight equal to the average of its two members' worker-level weights, and that pair weight enters all downstream regressions. This reweighting is orthogonal to the MEPS sampling weights, which target population representativeness within a year; a robustness specification that additionally incorporates the MEPS person weights yields nearly identical estimates (Table 9).

Two-part hurdle estimating equations

Both parts of the two-part hurdle described in Section 4.5 use the event-study form of equation (1), with the same matched-pair weighting and person-clustered standard errors, but on different samples.

Stage 1A. With $Z_{it} = \mathbb{1}\{\text{premium}_{it} > 0\}$, Stage 1A estimates

$$\begin{aligned} Z_{it} = & \mu_i + \alpha_t + \sum_{k \neq -6} \sum_e \lambda_k^e D_{ikt} \mathbb{1}\{e_i = e\} \\ & + \sum_{k \neq -6} \sum_e \beta_k^{1A,e} D_{ikt} T_i \mathbb{1}\{e_i = e\} + \varepsilon_{it}, \end{aligned} \quad (9)$$

on the full matched panel of $N = 43,989$ person-months. The eras $e \in \{\text{Pre}, \text{GR}, \text{Post}\}$ and the event-time bins k follow the definitions in Section 4.4.

Stage 2. Stage 2 estimates the burden ratio (and the high-burden indicator) conditional on a positive premium and household income at or above the floor:

$$\begin{aligned} \text{burden}_{it} = & \mu_i + \alpha_t + \sum_{k \neq -6} \sum_e \lambda_k^e D_{ikt} \mathbb{1}\{e_i = e\} \\ & + \sum_{k \neq -6} \sum_e \beta_k^{2,e} D_{ikt} T_i \mathbb{1}\{e_i = e\} + \varepsilon_{it}, \end{aligned} \quad (10)$$

on the within-payer subsample $\{(i, t) : \text{premium}_{it} > 0, Y_{it}^{\text{HH}} \geq \underline{Y}\}$, $N = 14,097$ person-months on 1,663 workers. The same specification is also estimated with the high-burden indicator $\mathbb{1}\{\text{burden}_{it} > 0.10\}$ as the dependent variable on the same Stage 2 sample.

Stage 1B: composition isolation among the privately covered

The two-part decomposition described in Section 3.3 reports two variants of the extensive premium-paying margin. *Stage 1A* (equation (9)) estimates $\Pr(\text{OOP} > 0)$ on the full matched panel and is reported in the headline Table 3; it captures both the extensive coverage margin (whether the

worker has private coverage at all) and the within-covered composition margin (whether the covered worker is on a premium-bearing plan). *Stage 1B* re-estimates the same indicator on the subsample restricted to person-months with observed private coverage, netting out the extensive coverage margin and isolating the composition shift among the privately covered.

The Stage 1B sample contains 21,134 person-months across 2,230 workers. Estimating equation (9) on this subsample, with coefficients renamed $\beta_k^{1B,e}$, yields the per-bin coefficients reported in Table 11. The pre-ACA peak is -15.1pp at the two-month horizon and -21.3pp at the four-month horizon (both significant at 1%); the post-ACA peak is -3.9pp at two months (not significant) and -10.8pp at four months (significant at 10%). The implied Stage 1B ACA mitigation is $+11.1\text{pp}$ at $k = +2$ ($p < 0.05$) and $+10.6\text{pp}$ at $k = +4$ ($p < 0.10$); the Stage 1B mitigation is larger than the Stage 1A mitigation reported in Table 3 ($+5.6\text{pp}$ at $k = +2$, $+8.2\text{pp}$ at $k = +4$).

The Stage 1B finding has a clean substantive interpretation. Among workers who retain private coverage post-displacement, the post-ACA cohort is markedly more likely to be on a premium-bearing plan than the pre-ACA cohort. This is direct evidence of non-employer private plans, predominantly Marketplace plans (which by construction require nonzero worker premium contributions), substituting for the employer-pays-100% ESI arrangements that more displaced workers held under the pre-ACA regime. The Stage 1B mitigation exceeds the Stage 1A mitigation because Stage 1A averages across workers who lose coverage entirely (those workers contribute zeros to both pre- and post-ACA bins, attenuating the extensive-margin pre/post difference) while Stage 1B conditions on continued private coverage and so picks up the composition margin alone. Read jointly, the Stage 1A and Stage 1B results indicate that the ACA expanded post-displacement coverage by a combination of extensive-margin retention (more workers staying covered) and within-covered composition shift (covered workers moving onto premium-bearing plans), with the composition shift the larger of the two margins.

The substantive Stage 2 finding documented in Section 5.1 should be read alongside this Stage 1B composition shift: the post-ACA cohort is moving onto premium-bearing plans (Stage 1B positive), and within those premium-bearing plans the displacement-induced burden share rises (Stage 2 positive). The two effects together produce the joint reading: Marketplace coverage is real insurance that substitutes for the employer-paid arrangements lost at displacement, but it imposes meaningful within-payer cost burden on the workers who take it up.

Table 11: Stage 1B event study: $\Pr(\text{OOP} > 0)$ conditional on private coverage

Era	$k = -4$	$k = -2$	$k = +0$	$k = +2$	$k = +4$
Pre-ACA	-0.043* (0.024)	-0.012 (0.024)	-0.118*** (0.030)	-0.151*** (0.040)	-0.213*** (0.044)
Post-ACA	-0.011 (0.039)	+0.005 (0.039)	-0.024 (0.045)	-0.039 (0.054)	-0.108* (0.057)
Mitigation	+0.032 (0.046)	+0.017 (0.046)	+0.094* (0.054)	+0.111** (0.067)	+0.106* (0.072)

Notes. Stage 1B re-estimates equation (9) on the private-coverage subsample (21,134 person-months, 2,230 workers), with coefficients renamed $\beta_k^{1B,e}$. The Stage 1B mitigation exceeds the Stage 1A mitigation reported in Table 3 (Stage 1A: +5.6pp at $k = +2$, +8.2pp at $k = +4$) because Stage 1A averages over workers who lose coverage entirely while Stage 1B conditions on continued private coverage. Standard errors clustered at the person level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Single-stage burden specification: reconciliation

The single-stage burden specification estimates equation (10) on the full *burden-valid* subsample, defined as the set of person-months with private coverage and household income at or above the \$1,000-per-year floor in 2016 dollars, *without* the additional restriction to positive out-of-pocket premium that Stage 2 imposes. On this subsample, burden_{it} equals zero by construction for employer-pays-100% person-months and equals the realized premium share for workers who do pay any premium. The estimator therefore averages over two distinct populations whose burden values are generated by different mechanisms.

Because Stage 1A is positive (post-ACA workers are more likely to land on a premium-paying plan after displacement) and Stage 2 high-burden is positive (conditional burden among payers rises post-ACA), the single-stage estimator combines a composition shift onto premium-paying plans with an intensity effect on the burden share, the two of which move in opposite directions. The single-stage mitigation is therefore close to zero by construction. This is a reconciliation, not a contradiction: the two-part decomposition is the specification that recovers the two effects separately, and the near-zero single-stage estimate is the object the decomposition explains.

Comparison with [Goldman et al. \(2018\)](#)

The Stage 2 specification departs from the cross-sectional difference-in-differences in [Goldman et al. \(2018\)](#) on three margins, each dictated by the displacement-design context. First, on the numerator: their numerator is total out-of-pocket health spending (premiums plus point-of-service costs); mine is the out-of-pocket premium only. The relevant question for a displaced worker is the price of remaining covered, not the unrelated cost of the year's specific medical events. Restricting the numerator to premiums isolates the affordability margin the ACA's subsidy provisions actually target. Premiums and point-of-service spending move on different margins post-ACA: [Goldman et al. \(2018\)](#) document that mean premium contributions rose primarily in the non-subsidy-eligible

high-income group ($> 400\%$ of FPL), while out-of-pocket point-of-service spending fell across the lower-income groups eligible for subsidies and Medicaid expansion. A single combined numerator would therefore average across two distinct shifts on two distinct populations. Second, on the functional form: their dependent variable is log spending; mine is the level burden ratio (and the level-threshold indicator), following the burden-literature convention from [Selden \(2009\)](#) onwards. As a robustness check, I re-estimate Stage 2 with $\log(\text{burden}_{it})$ on the same within-payer subsample; the qualitative pattern is the same, with the displacement-induced log-burden rise larger post-ACA than pre-ACA and mitigation contrasts of comparable significance to the level-form headline. Third, on the design: their specification is a cross-sectional difference-in-differences with one pre period and one post period, recovering a single average treatment effect. The matched event-study design here recovers a dynamic profile across nine event-time bins and three eras, with the matched-pair reweighting and inverse-odds weights described in [Section 4.2](#). The cross-sectional setup is appropriate for the population-level coverage and spending evaluation that is their objective; the dynamic event-study setup is necessary for the displacement-specific analysis, which requires identifying the event-relative trajectory.

C Matching diagnostics, mitigation contrast, and pre-trends

This appendix supports the matching procedure described in [Section 4.2](#) and the mitigation contrast standard error referenced in [Section 4.4](#).

Score-distribution balance

Figures [18](#) and [19](#) show the propensity-score distributions before and after matching, separately for treated and control. Pre-matching, the distributions overlap throughout the support but treated workers concentrate more in the right tail. Post-matching, the two distributions are near-identical, confirming successful score balance within cells. Standardized mean differences on every covariate in $\mathbf{W}_i$ fall below 0.10 after matching, the conventional threshold for adequate covariate balance in applied propensity-score work ([Austin, 2011](#)), justified by the central role of the propensity score in balancing the full covariate vector ([Rosenbaum and Rubin, 1983](#)). Balance tables are available from the author on request.

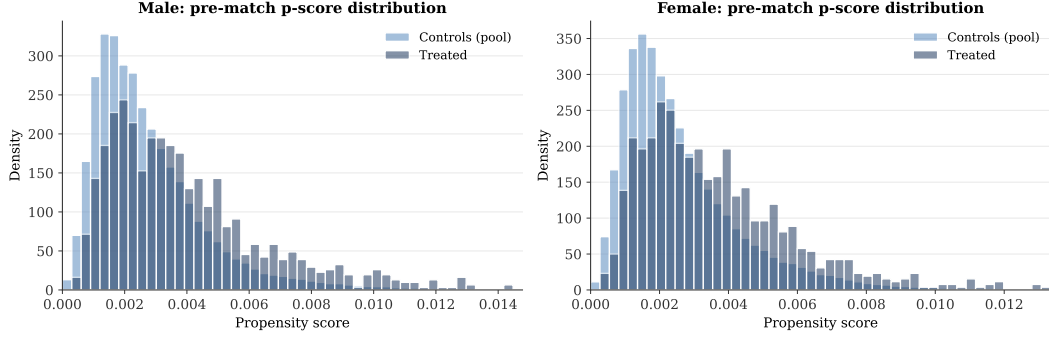


Figure 18: Propensity score distribution: pre-matching

Density of estimated propensity scores before greedy nearest-neighbor matching.

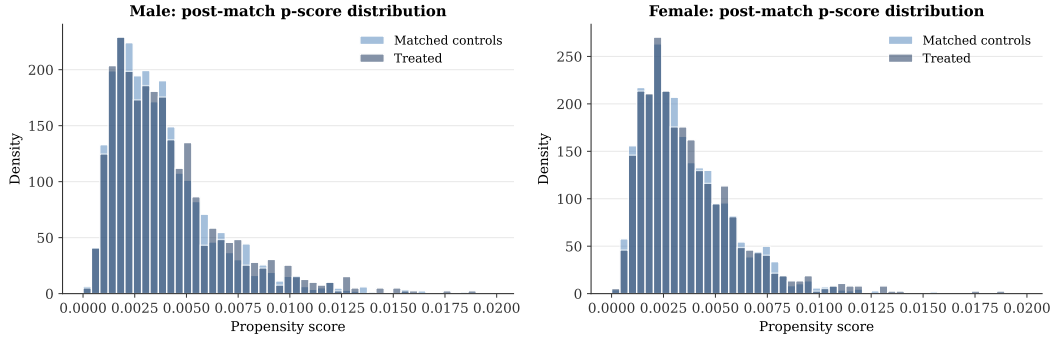


Figure 19: Propensity score distribution: post-matching

Density of estimated propensity scores after greedy 1:1 nearest-neighbor matching within exact cells. Treated and control distributions are visually indistinguishable.

Mitigation contrast standard error

The standard error of the cross-era contrast $\hat{\Delta}_k$ defined in equation (2) is computed from the full estimated variance-covariance matrix $\hat{V}$ of the event-study coefficients in equation (1):

$$\widehat{\text{SE}}(\hat{\Delta}_k) = \sqrt{\hat{V}_{kk}^{\text{Pre,Pre}} + \hat{V}_{kk}^{\text{Post,Post}} - 2\hat{V}_{kk}^{\text{Pre,Post}}}. \quad (11)$$

Each worker is in precisely one era throughout the event window, so the pre-ACA and post-ACA workers form disjoint clusters and the cross-era covariance term $\hat{V}_{kk}^{\text{Pre,Post}}$ is identically zero; equation (11) reduces to the independent-variance form $\sqrt{\hat{V}_{kk}^{\text{Pre,Pre}} + \hat{V}_{kk}^{\text{Post,Post}}}$. The same construction applies to the two-part contrasts $\hat{\Delta}_k^{1A}$ and $\hat{\Delta}_k^2$ in equation (3), with $\hat{V}$ taken from the corresponding stage-specific regression.

The cross-era contrast carries a power cost relative to the era-specific coefficients: the standard error of a difference of two coefficients exceeds the standard error of either coefficient individually, so $\hat{\Delta}_k$ can fail to reach conventional significance even when both $\hat{\beta}_k^{\text{Pre}}$ and $\hat{\beta}_k^{\text{Post}}$ are individually

highly significant. This is most visible in the stratified specifications, such as the income-bin breakdown reported in Table 8, where within-bin sample sizes are small.

Pre-trends and the 2014 break

The matched event-study design relies on parallel pre-displacement trajectories within each era and, for the cross-era contrast, on symmetric pre-trends across eras. This subsection reports formal tests of both.

For each event-study specification estimated in the paper, I compute three joint Wald tests on the pre-displacement event-time leads ($k = -4$ and $k = -2$):

- **Test (i)** (pre-ACA pre-trend): $H_0 : \beta_{-4}^{\text{Pre}} = \beta_{-2}^{\text{Pre}} = 0$.
- **Test (ii)** (post-ACA pre-trend): $H_0 : \beta_{-4}^{\text{Post}} = \beta_{-2}^{\text{Post}} = 0$.
- **Test (iii)** (mitigation contrast pre-trend): $H_0 : (\beta_{-4}^{\text{Post}} - \beta_{-4}^{\text{Pre}}) = (\beta_{-2}^{\text{Post}} - \beta_{-2}^{\text{Pre}}) = 0$.

Test (iii) is the substantive test for this paper’s design. The reason is that the cross-era mitigation contrast $\hat{\Delta}_k = \hat{\beta}_k^{\text{Post}} - \hat{\beta}_k^{\text{Pre}}$ remains identified even when (i) or (ii) individually reject, provided the pre-trends are symmetric across eras and difference out at the leads. A specification can therefore fail (i) or (ii) yet pass (iii), and only the latter bears on the causal interpretation of the mitigation contrast.

Across 59 specifications spanning the main coverage trio, the two-part hurdle, the employment and earnings validation, the income-band stratification, the per-class and super-class LCA event studies, and the HQ-vs-LQ DDD, test (iii) fails to reject at the 5% level in all 59. Three specifications are near the 10% level — two below (the C2 Trade and Sales any-insurance contrast at $p = 0.082$ and the super-class Precarious Stage 2 burden-ratio contrast at $p = 0.092$) and one just above (the C2 public-coverage contrast at $p = 0.103$); these are disclosed in the relevant results subsections. Ten specifications fail (i) or (ii) individually while passing (iii) cleanly. The Stage 1A full-panel specification is representative of this pattern, with pre-ACA $p = 0.028$, post-ACA $p = 0.317$, and mitigation contrast $p = 0.965$, indicating symmetric pre-trends across eras whose displacement effects are identifiable from the era contrast even though the pre-ACA leads are individually nonzero. These specifications are not flags for the mitigation interpretation.

The 2014 break is also visible at the panel level. Figure 21 plots MEPS panel-level estimates of $\hat{\beta}_{+2}$ on any insurance, with each panel placed at its median calendar year. Pre-2014 panels cluster around -0.22 to -0.33 , post-2014 panels drop sharply, and the implied structural break at the 2014 implementation date is clean: it is not consistent with a smooth secular trend that would produce a graduated decline across panels.

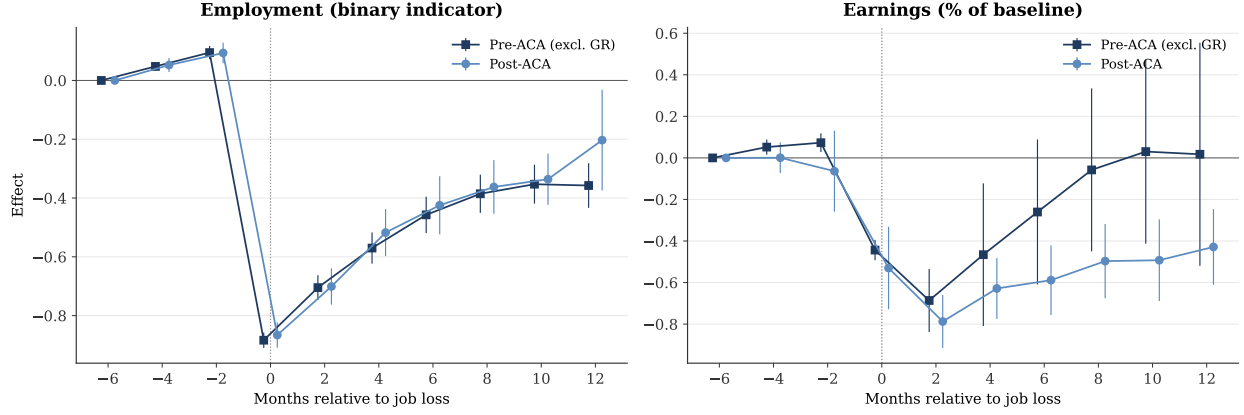


Figure 20: Event-study validation: pre-trend tests across outcomes

Validation of pre-trends by plotting $\hat{\beta}_k$ at $k \in \{-4, -2\}$ across the headline outcomes in both eras, with 95% confidence intervals. All coefficients are individually small and statistically indistinguishable from zero at $k = -4$ for the headline insurance outcome, confirming parallel pre-trends.

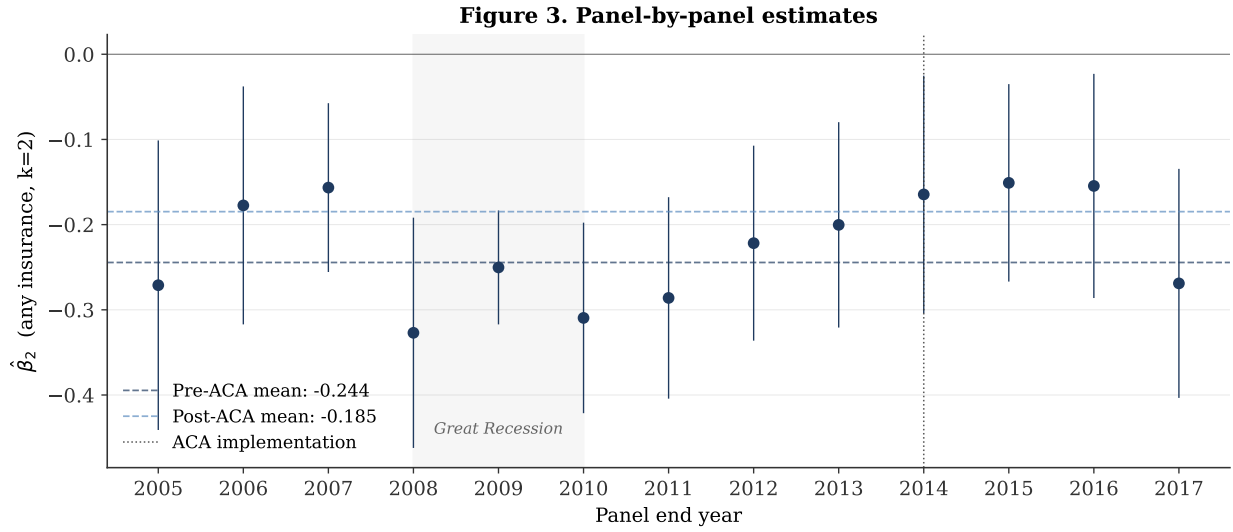


Figure 21: Panel-level estimates of any-insurance peak loss

MEPS panel-level estimates of $\hat{\beta}_{+2}$ on any insurance, plotted by the panel's median calendar year. Pre-2014 panels cluster around -0.22 to -0.33 ; post-2014 panels drop sharply. The structural break at 2014 is visible and is not driven by a secular linear trend.

D Latent class analysis details

This appendix supports the latent class analysis described in Section 4.6.

Indicator construction with explicit cutoffs

The 23 binary indicators measured at $k = -6$ are: thirteen job-quality features, six industry dummies, and four occupation-group dummies. The job-quality indicators and their exact cutoffs are:

- **Full-time:** usual hours ≥ 35 per week.
- **Long hours:** monthly hours > 180 (about 45 hours per week).
- **Multiple jobs:** respondent reports more than one active job in the round.
- **High wage:** monthly earnings above the sample 75th percentile (\$2,993 in 2016 dollars on the analytical sample).
- **Low wage:** monthly earnings below the sample 25th percentile (\$1,077 in 2016 dollars).
- **Employer offer:** employer offers health insurance.
- **Paid leave:** worker has paid sick leave or paid medical leave.
- **Vacation:** worker has paid vacation.
- **Pension:** worker has employer-sponsored pension or retirement plan.
- **Union:** worker is a union member.
- **Large employer:** firm size ≥ 50 employees (the ACA employer-mandate threshold).
- **Temporary:** worker reports the job is temporary.
- **Seasonal:** worker reports the job is seasonal.

Industry dummies merge MEPS categories into six groups (Trade/Transport, Finance/Professional, Health/Education, Service, Government, and *Other*), with *Goods* (Agriculture, Construction, Manufacturing) as the omitted reference. Occupation dummies merge into four groups (Sales/Office; Blue-collar; Service; *Other*), with *Professional/Management* as the omitted reference.

Listwise deletion accounting

Starting from the 3,709-person matched pool, the analytical sample applies (i) drop Military industry/occupation (~ 22 persons), (ii) listwise drop on any of the 23 indicators (~ 22 persons), yielding 3,665 persons (1,875 treated, 1,790 controls). The total drop is under 2%.

***K*-selection criteria**

Following the multi-criterion logic of [Brauner et al. \(2019\)](#), K is selected jointly on information criteria (Bayesian information criterion BIC, sample-size adjusted BIC), classification quality (entropy and minimum average posterior class probability AvePP), minimum class size, and substantive interpretability of the class profiles, rather than minimizing BIC alone. As Table 5 shows, information criteria fall monotonically through $K = 7$, but classification quality and minimum class size deteriorate sharply at $K \geq 5$. I select $K = 4$, the largest K at which all criteria are jointly satisfied. At $K \geq 5$, the smallest classes shrink to 3.5% (at $K = 5$) and 3.4% (at $K = 6$) of the analytical sample, close to the small-class threshold [Brauner et al. \(2019\)](#) apply in their own LCA (where they reject a class below 2% on generalizability grounds) and consistent with the class-collapsing pattern [Masyn \(2013\)](#) identifies as a sign of overextraction. By contrast, $K = 4$ retains entropy 0.906 and minimum AvePP 0.917, both well above conventional thresholds, and yields the clearest substantive class identities.

Class profiles, all four classes

The defining contrasts across the four classes are:

- **C3 (High-quality)** differs from the other three on nearly every job-quality indicator: employer offer 97% vs. 38–49%; pension 70% vs. 4–9%; vacation 97% vs. 20–28%; paid leave 88% vs. 8–13%; full-time 95% vs. 55–83%.
- **C1 (Service)** is the most distinctly Service-sector class: 87% Service occupation vs. 0–16% in the other classes; lowest full-time rate (55%) and lowest pension rate (4%).
- **C2 (Trade and Sales)** concentrates in retail/sales: Trade/Transport industry (32%) and Sales/Office occupation (61%); highest temporary-job rate (16%); intermediate employer offer (49%) but very low benefit coverage.
- **C4 (Goods and Blue-collar)** is the manufacturing/construction class: Goods-producing industry 64% (the omitted reference) and Blue-collar occupation 100%; high full-time rate (83%); low employer offer (46%) but moderate vacation and pension coverage.

Largest-gap rule numerics on this fit

The four classes' baseline employer-coverage offer probabilities are $\pi_{\text{offer}} = 0.383$ (C1), 0.491 (C2), 0.973 (C3), and 0.460 (C4). Sorting yields $\{0.383, 0.460, 0.491, 0.973\}$. The gap structure between consecutive sorted values is 0.077, 0.031, 0.482. The single largest gap (0.482, between 0.491 and 0.973) defines the Good-job/Precarious split, with C3 alone in the Good-job tier.

Demographic composition of the four classes

The latent class structure is estimated from the 23 baseline job-quality and industry/occupation indicators only; demographics (age, sex, race, marital status) do not enter the LCA likelihood. As a descriptive check, Table 12 reports class-level demographic means on the LCA analytical sample. The table is descriptive only and is not used for any inference; its role is to characterize who the four classes are demographically, given that the LCA itself recovers them from job-quality indicators alone.

The demographic patterns are consistent with the substantive class identities. C1 (Service) and C2 (Trade and Sales) skew female (66% and 64% respectively) and slightly younger (mean age ≈ 38.9). C3 (Good-job) is gender-balanced (49% female), oldest (mean age 41.0), and the most likely to be married (54%). C4 (Goods and Blue-collar) is heavily male (24% female). Hispanic representation is highest in C1 (37%) and C4 (42%) and lowest in C3 (20%); Black non-Hispanic representation is somewhat higher in C1 (27%) than in the other three classes (19–22%). Education sorts along the job-quality axis: C3 has the highest college-completion share (29%) and the lowest below-high-school share (9%), while C4 has the inverse pattern (3% college, 40% below high school), reflecting that C4 contains skilled-trades workers who entered without college credentials. C1 and C2 are intermediate; C2 has roughly twice the college share of C1 (17% vs. 7%). The baseline poverty distribution mirrors the education ranking: C3 has only 6% in poverty and 37% at high incomes ($\geq 400\%$ FPL), while C1 has 35% in poverty and just 6% at high incomes. Poverty also concentrates in the income band where the ACA's premium subsidies are largest: in C1, 35% of workers fall in the 100–199% FPL bin at baseline, the band where Section 5.3 finds the largest mitigation effect. The cross-class differences are descriptive associations rather than independent predictors of class membership; the LCA classes are recovered from job characteristics, and the demographic table reports the populations that those classes happen to contain.

Table 12: Demographic composition of the four LCA classes

	C1 (Service)	C2 (Trade and Sales)	C3 (Good-job)	C4 (Goods/Blue-collar)
Age (years)	38.9	38.9	41.0	39.7
Female (%)	66	64	49	24
Married (%)	45	46	54	53
Black, non-Hispanic (%)	27	22	22	19
Hispanic (%)	37	26	20	42
Asian, non-Hispanic (%)	4	7	7	5
Below high school (%)	30.6	13.9	8.8	39.8
High school only (%)	51.2	51.6	44.9	48.9
Some college (%)	11.2	17.7	17.2	8.2
College or more (%)	7.1	16.7	29.1	3.1
Poor, < 100% FPL (%)	34.5	23.2	5.6	19.7
Low Income, 100–199% (%)	35.4	29.8	17.1	35.9
Middle Income, 200–399% (%)	23.9	29.3	40.1	33.5
High Income, ≥ 400% (%)	6.2	17.7	37.1	10.9

Notes. Class-level means on the LCA analytical sample (3,665 baseline-employed persons; 1,875 treated, 1,790 controls). Education codes: “below high school” is code 1, “high school only” is code 2 (high school graduate, no college), “some college” is code 3 (some college, no degree), “college or more” is code 4 or 5 (Bachelor’s, Master’s, Doctorate, or professional). The four education shares sum to 100% within each class. Poverty bins follow the four-bin merger used in Section 5.3 and Table 8: “Poor” is family income below 100% of the federal poverty level (FPL); “Low Income” merges the original MEPS Near-Poor (100–124%) and Low Income (125–199%) bands into a single 100–199% FPL bin to match the paper’s main income heterogeneity specification; “Middle Income” is 200–399% FPL; “High Income” is ≥ 400% FPL. Poverty shares sum to 100% within each class. The LCA is estimated on the 23 baseline job-quality and industry/occupation indicators only (Section 4.6); demographics do not enter the LCA likelihood. The table is descriptive only.

E Statement on AI use

In preparing this paper I used Anthropic’s Claude (Claude Sonnet 4.5 and Claude Opus 4.7) as a coding and writing assistant. The model helped (a) structure and debug the Stata build pipeline and the Python event-study and latent-class notebooks, including specifications, diagnostics, and figure-formatting code; (b) draft, revise, and proofread sections of the manuscript, including expository sentences in the empirical-strategy, results, and limitations sections; and (c) prepare the LaTeX formatting (figure and table captions, cross-reference structure, and bibliographic formatting). All identification choices, research-design decisions, methodological choices, the substantive interpretations including the data-handling decisions and the final wording of every sentence

were made by me. I have personally verified all numerical results reported in the paper against the executed Jupyter notebooks. Any remaining errors are mine.